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Multi-dimensional View of Microplastic Toxicity: Environmental and Health impacts

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Abstract MPs are pervasive environmental pollutants prevailing in aquatic, terrestrial environment and the surrounding atmosphere. MPs are of global concern due to their discharge into the environment and adverse effect on entire biota. Due to their abundance, MPs pose a risk to organisms and environmental health. Therefore, an alleviation of MP pollution is a problem to be approached via one health perspective. The review compiles recent literature on health risks that MPs pose along with rising ecotoxicological impacts particularly atmospheric, aquatic and terrestrial. Additionally, an attempt is made to connect the rising incidence of non-communicable diseases with human consumption MPs. Adverse effects ranging from biogeochemical cycling to aquatic biota to terrestrial impacts are viewed through a one health lens and solutions proposed. The routes of MP into the food chain, toxicity through bioaccumulation or trophic transfer are also reviewed. The chronology of MP accumulation to recent literature on in vitro cell culture and investigations using model organism platforms have been teased to investigate the adverse impact of MPs on different organ systems of humans leading to acute and/or chronic toxicity. Another global pandemic due to MP distress appears underway. A complex issue as this can be tackled by scientists, public health experts, environmentalists and common men working in concert.

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1 Introduction

Plastics are a anthropogenic menace; the manufacture and consumption of which is on rise dramatically since 1950’s which was 2 million tonnes (mt) (Prata et al., 2021; Raza et al., 2022). According to Organization for Economic Co-operation and Development (OECD) global plastic outlook (2022), the global plastic production was 459.75 mt in 2019 and will further reach to 975.63 mt by 2050 (Geyer et al., 2017). And India’s plastic consumption grew 23-fold between 1990 and 2021, reaching almost 21 million tons. After the United States and China, India is the third-largest user of plastics, accounting for around 6% of the world’s total usage as reported by UJA market report (2023). Plastics offer durability, waterproofness, low cost and light weight. However, they are recalcitrant and non-biodegradable. Based on shape and size, the environmental plastic litter can be categorized into four types; macroplastics (> 25 mm), mesoplastics (5–25 mm), microplastics (1 µm-5 mm) and nanoplastics (1–1000 nm and < 20 µm).

Microplastics (MPs) are water insoluble synthetic polymers of less than 5 mm in size derived from plastics. The United Nations Environmental Programme (UNEP) Year Book 2014 lists MPs as one of the ten emerging pollutants that could endanger the health of people and other animals in all biomes (Constant et al., 2020). MPs differ in type, shape and chemical properties in the environment. Fibers are the most abundant shape in the environment (Avinash et al., 2023). Also, MPs are of several types based on their chemical composition. Polypropylene (PP 19.3%), low-density polyethylene (LDPE 17.5%), high-density polyethylene (HDPE 12.3%), polyvinyl chloride (PVC 10.2%), and polyurethane (PUR 7.7%), polystyrene (PS 6.6%), and polyethylene terephthalate (PET 7.4%) are the most widely produced polymer kinds (Petersen & Hubbart, 2021; PlasticsEurope, 2019). (Ali et al., 2024; Islam & Cheng, 2024) reported high prevalence of small particle sized (less than 500 µm) and polyethylene (PE), PP, and PET polymers in the environment. The global plastic footprint (plastic production, waste generation and leakage into the terrestrial aquatic environment (river is contributing to ocean and to the human) is depicted in the Fig. 1.

The existence of tiny plastic particles has been reported in the oceans as early as the 1970s. However, a groundbreaking study headed by marine ecologist Richard Thompson in 2004 marked the official start of research into the plastic distribution and effects. Further, more investigations revealed an unexpectedly high percentage of plastic contamination. Apart from marine system, freshwater ecosystem received attention with the first study published in 2011 (Zbyszewski & Corcoran, 2011). These freshwater MPs eventually end up in the ocean. According to (Jambeck et al., 2015) 4.8 to 12.7 mt enters ocean yearly. Due to salinity of the ocean; degradation does not occur as well as it does in freshwater. MPs can travel via water, air, and soil. And, since the hydrosphere, atmosphere, and pedosphere are interconnected in the ecosphere; they are omnipresent. Land-based sources account for 80% of MP particle with the remaining 20% coming from marine sources (Belioka & Achilias, 2024; Rochman, 2018). It was reported that between 1950 and 2015, 4.97 billion tonnes of plastic waste were dumped in landfills and open environment (Geyer et al., 2017). (Shen et al., 2020) convey that 94% of the 320 mt of plastics generated yearly, ended up in the surrounding ecosystem. MPs accumulate in aquatic and terrestrial biota due to their small size and transfer to higher trophic levels including humans via food chain referred to as 'biomagnification'. Human exposure to MPs may result from an extensive utilization of plastics, environmental pollution especially post-use plastic exposure via eating contaminated food. An approximate consumption of 39,000–52,000 MP particles/person, yearly has been reported (Cox et al., 2019).

MPs are ingested and adhere to aquatic organisms, including planktons, aquatic plants, fish, and turtle and exert toxicological effects (Elgarahy et al., 2021; Tian et al., 2023). MPs exert negative effect due to (1) their adsorption ability (greater surface area and hydrophobicity), they adsorb other pollutants such as persistent organic pollutants (POPs) and heavy metals (HMs) and thus serving as a vector and (2) them containing additives viz. plasticizers, colorants, heat stabilizers, flame retardants etc. that leach into the environment (Ali et al., 2024). Additionally, co-exposure to microplastics and other pollutants can worsen toxicity and disrupt ecological equilibrium.

After entering the soil, these particles can build up, and eventually reach concentrations that may adversely impact the biodiversity and soil function, and consequently, terrestrial ecosystems (Rillig, 2012). Also, MP affect soil properties, nutrient storage, cycling and greenhouse gas (GHG) emissions (Kumar et al., 2023). MPs consumption can cause serious health disorders that can be life threatening.

In the graph (Fig. 2B), the rising global plastic production (Ritchie et al., 2023) concomitant with an increasing global burden of non-communicable diseases (NCDs) in response through the years 2010 to 2020 is represented. Figure 2A compiles the data on NCDs (Roser et al., 2024) which were prevalent in 2017 and 2019. These were cardiovascular diseases, neonatal disorder, diabetes, cancer, kidney disorders and, respiratory diseases. The consumption of MPs and their accumulation may have contribution towards NCDs.

The adverse effects of MP pollution on the environment must be viewed from a broader perspective of one health approach starting from terrestrial, aquatic ecosystem until it reaches humans given that all are interlinked. However, it is crucial to emphasize that the ecotoxicological effects and mechanisms of action of MPs and pollutants adsorbed to MPs in organisms are still unknown. To address the gaps in MP research; we conducted a thorough and

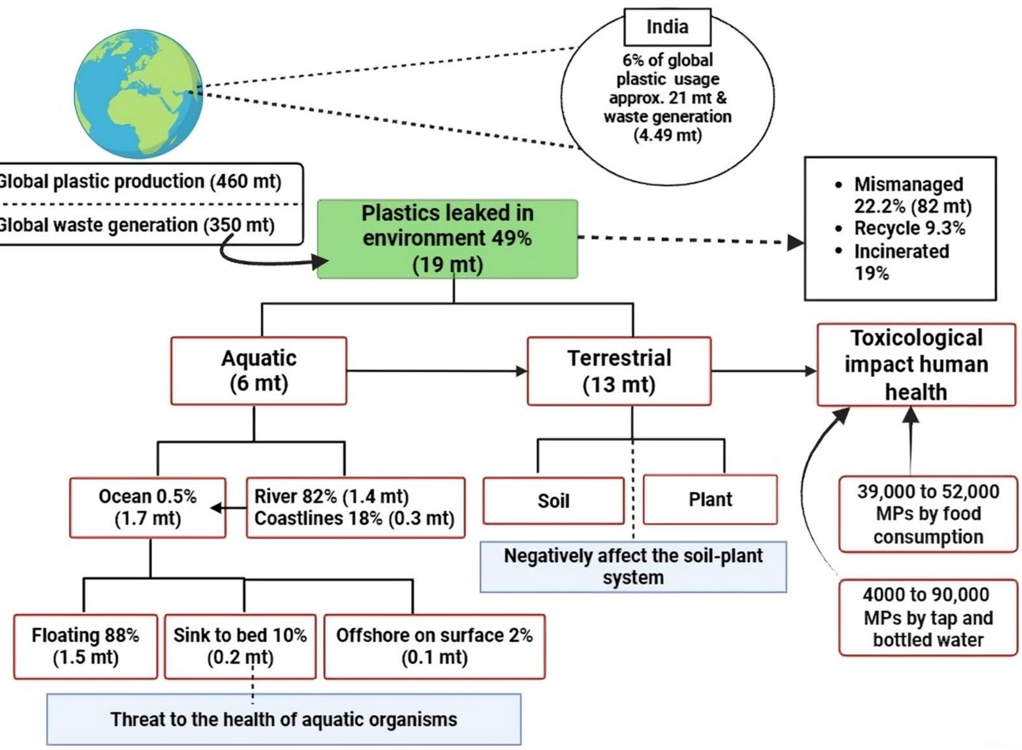


Fig. 1 Global plastic footprint and its distribution (in mt) into the different environmental compartments

comprehensive evaluation of the literature on microplastics. The objectives of this review are (1) to provide an overview of the occurrence, abundance and fate of MPs in the entire ecosystem, (2) to examine their possible toxicity responses on animal and human health, with the possible mechanism involved, (3) to elucidate the MPs effect with co-contaminant. Because of the multidimensionality of the MP menace; it is also proposed that the resolution of microplastic toxicity is not a one-way approach but requires multi-disciplinary approaches to alleviating them.

2 Toxic Biotic and Abiotic Impact of Microplastics on Environmental Health

MP pollution is currently regarded as a developing threat to ecosystem function and biodiversity. MPs get accumulated in soil, impact the soil properties and afflict the soil inhabiting organisms. The contaminated soil is taken up by the plants and impacts the plant health Fig. 3.

2.1 Microplastics Impact on Terrestrial Soil and Plant Ecosystem

Currently, the main contributing element to soil MP pollution is the widespread use of agricultural films in agroecosystem. MPs in soil change the soil chemical properties such as pH, conductivity, nutrient cycling (carbon, nitrogen content), dissolved organic matter (DOM), cation exchange capacity (CEC) and, exopolymeric substances (EPS) (Ma et al., 2023). They impact the physical properties such as aggregation (weakening the structure and reducing soil stability), bulk density, porosity, and water holding capacity (de Souza Machado et al., 2019; Feng et al., 2022). MPs have also been reported to harm the soil microbial community and soil fauna which enable breaking down of organic matter and release of nutrients (Qian et al., 2018; Ya et al., 2022). MPs also affect the soil enzyme activities like catalase (CAT), urease, and dehydrogenase (DHA) due to their strong adsorption capacity there by affecting the soil fertility and nutrients (Fei et al., 2020; Feng et al., 2022). (Irshad

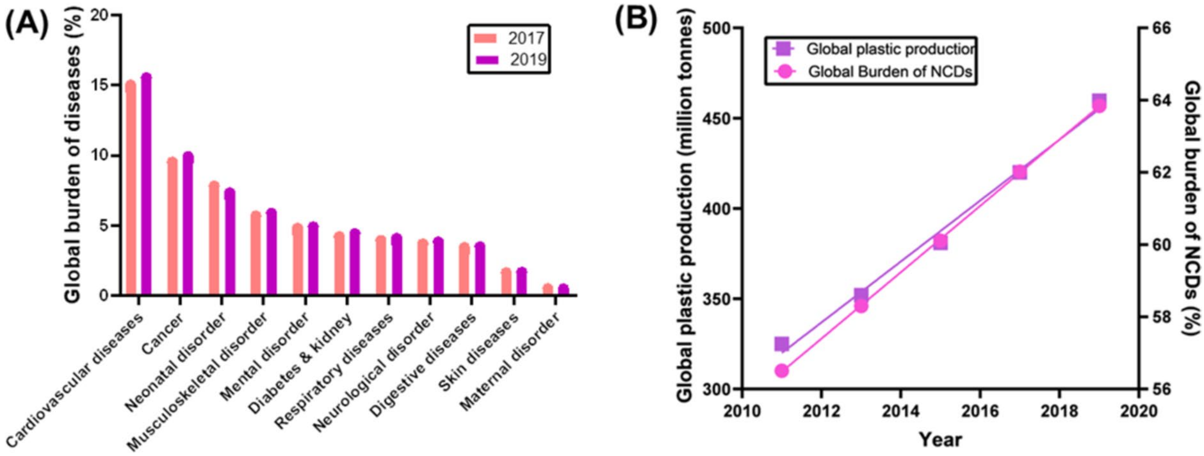


Fig. 2 Graph showing the relationship between plastic production and NCDs

et al., 2024) reported an increase in soil enzymatic activities and an altered rice-rhizosphere bacterial community due to polylactic acid (PLA) MPs. Additionally, MP pollution of water and soil ecosystems impacts the biogeochemical cycling, which is crucial for the ecological stability of terrestrial environment. Recent experimental research suggests that, MPs may raise emissions of greenhouse gases (GHGs), including nitrous oxide (N_2O), carbon dioxide (CO_2), and methane (CH_4) and also the C,N, and P cycles in soil (Lan et al., 2025a, b; Liu et al., 2024a, b, c; Zhang et al., 2022a, b, c, d, e, f). MPs alter community networks and structures, enzyme activity, and functional gene expressions linked to nutrients, which disrupts the ecological balance of microbially driven nutrient cycling (Chen et al., 2024a, 2024b, 2024c, 2024d; Miao et al., 2019; Yu et al., 2022). MPs alter the soil nitrogen cycling genes and thus aiding in soil nitrogen mineralization and transformation (Zhou et al., 2023a, 2023b). The aforementioned results offer compelling reasons to investigate the effects of MPs on soil microbial community functions related to biogeochemical cycling. Prevalent MPs'such as PA, PE, PP, PVC and PET (2%) impact the soil bacteriome with increased exposure duration. α-diversity index showed that soil volume surrounding MPs are enriched with microplastic adapted bacteria (Actinobacteria) while the abundance of *Bacteroidetes* and *Gemmatimonadetes* declined (Zhang et al., 2023a, 2023b, 2023c, 2023d, 2023e). (Sun et al., 2022a, 2022b) reported a strong correlation ($R^2 = 0.592$, $p < 0.001$) between altered concentration and composition of MPs mainly fiber, foam, and film and the community dissimilarity of soil bacteria. Recently, (Deng et al., 2024) revealed that the interaction between soil MPs and the surrounding bacteria diminished with age of MP due to changes in latter's hydrophobicity. Compared to the surrounding soil, the bacterial populations in plastisphere were different and less diverse. Similarity in microbial diversity of ARGs (antibiotic resistance genes) were more prevalent in agricultural soils from areas which are MP contaminated due to long-term use of plastic mulch (Lu & Chen, 2022; Wang et al., 2024; Wu et al., 2023). This is due to a horizontal transfer triggered by the mobile genetic elements (MGEs). Soil microbes and ARGs are regulated by soil metabolites driving antibiotic resistome through the metabolism-bacteria-ARGs linkage. An investigation on 3rhizosphere and MP amendment together on antibiotic resistome revealed that low amount of MPs have significant impact on ARG abundance. A significant increase in total abundance from 0.236 to 0.367 in bulk soil and a significant drop from 0.267 to 0.167 in rhizosphere soil was observed (Su et al., 2025). Higher amount of PE-MPs promoted high abundance of ARGs (tetracycline) and MGEs in rhizospheric soil compared to non-rhizospheric soil. MPs also promoted transfer of sulfa ARGs through soil from root to leaf of lettuce plant. Correlation and network analysis revealed that bacterial communities and MGEs were the main drivers of ARGs variation in soil-lettuce systems (N. Li

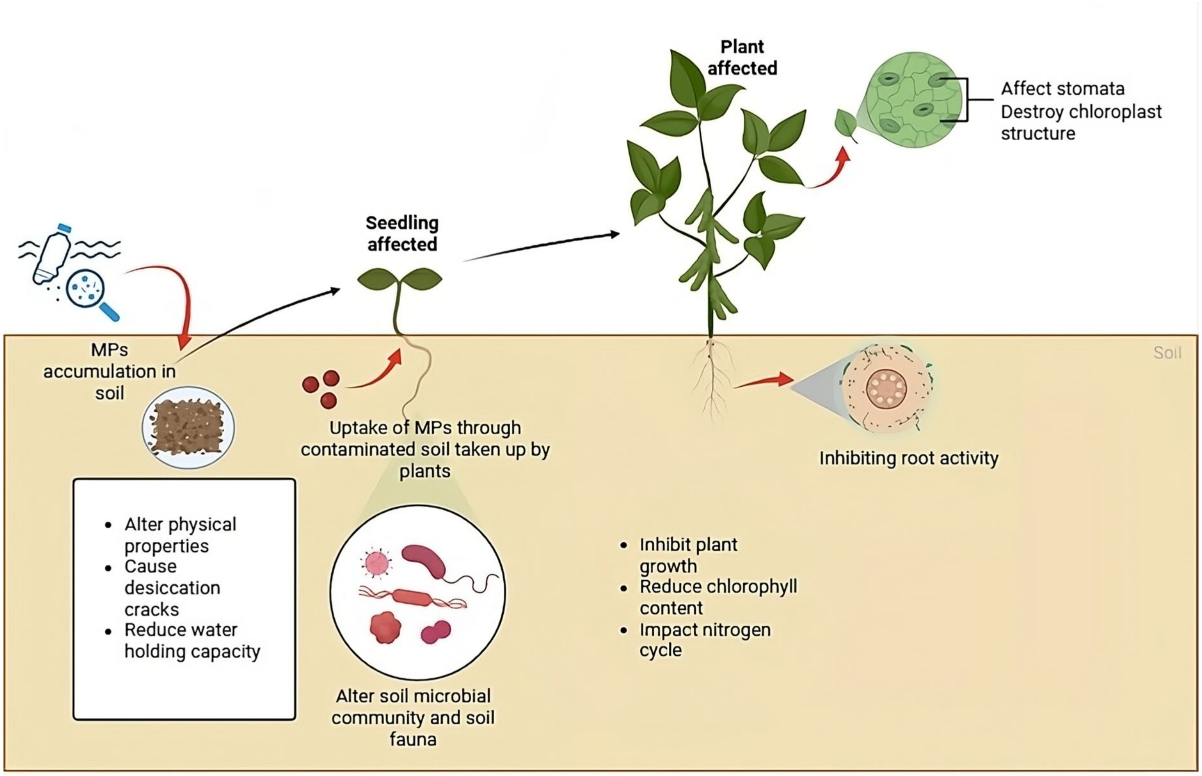


Fig. 3 Ecotoxicology of MP pollution on soil and plant ecosystems

et al., 2024a, 2024b, 2024c). The resulting poor soil quality may have repercussions on crop productivity which warrants elimination of entry of MPs into soil via irrigation water, agricultural biofilms etc.

Plants constitute the fundamental biotic element of the terrestrial ecosystem, therefore comprehending the interplay between MPs and plants is imperative. Endocytosis of MPs affects seed germination, plant growth and photosynthetic activity as reported in *Oryza sativa* (rice) (Wu et al., 2021a, 2021b); sweet potato (Khan et al., 2023) and; *Vigna radiata* (mung bean) (T.-Lee et al., 2022a, 2022b). (Saha et al., 2024) reported the presence of PET and HDPE mainly in the upper soil layers along with suppressed leaf area index and root growth. MPs took lesser time to reach plants if located in upper layer of soil and thus induced higher toxicity. *Cucurbita pepo* L. grown on contaminated soil with PP, PVC, PET, and PE in which PVC was found to be more toxic to shoot growth with reduction in leaf size and chlorophyll content of the plant (Colzi et al., 2022). It has been revealed that due to MPs surface charge and hydrophobicity; they adhere to the plant roots or seed surface and form hetero-aggregates of opposite charges. This blocks pores in the cell wall, inhibits water, nutrient absorption and respiration thus inhibiting the root and bud growth. Mask-derived MPs have been reported to reduce lettuce (*Lactuca sativa*) germination and showed hormesis effect (Song et al., 2024). MPs may adhere to the cell membrane and induce toxicity by destroying them. The reason for this is that the electrostatic attraction between PS-MPs particles and the cell membrane alters the aquaporin transport process and the membrane's hydrophobicity, which leads to an imbalance in the osmotic pressure inside and outside the membrane and, eventually, membrane rupture (Giorgetti et al., 2020). PS-MPs and polytetrafluoroethylene (PTFE) get adsorbed on the root surface via van der Waals interaction; thereby inhibiting root activity and restricting the nutrient uptake and decreases the biomass and yield. Adsorption of Arsenic (As(V)) by plants was also inhibited. Consequently, iron-reducing microorganisms' ability to use root exudates was suppressed in the presence of MPs. This was due to their strong interaction shown by molecular docking. Also observed was decreased the abundance of *Geobacteria* and *Anaeromyxobacter*. MPs lowered the hemoglobin content by damaging tertiary structure of rice non-symbiotic hemoglobin crystals (ID8U) thereby impacting the rice growth and yield. This was due to ROS in rice, and a raised the hemoglobin content (Dong et al., 2022). A hydroponic study conducted on sweet potato for 14 days revealed that PVC-MPs (100 and 200 mg/L) alone did not affect the plant physiological parameters but in combination with heavy metals chromium Cr(VI) (5, 10, and 20 µM) enhanced their toxicity. Thus, more adverse effects on physiological and biochemical properties were observed including leaf and root morphology, plant height, and plant biomass. Also, MPs enhanced Cr bioaccumulation in leaves, stem and roots with increased doses. This caused reduction in stomatal size as confirmed by the scanning electron microscope (SEM) thus hampering the photosynthetic and gas exchange functions. MPs + Cr induces oxidative stress (OS) and caused increased malondialdehyde (MDA) and hydrogen peroxide (H_2O_2) levels (Khan et al., 2023). The biological parameters and physiological responses of *Lepidium sativum* (garden cress) were affected by the exposure to PE, PVC, PP and their mixture induced OS in the plants. It has been observed that during acute exposure (6 days) H_2O_2 production is higher compared to chronic exposure (21 days) except in PVC-MPs. PVC induces higher toxicity compared to other MPs as it showed higher aminolaevulinic acid and proline levels (Pignattelli et al., 2020). Recently, (Wu et al., 2024) investigated toxicity of photodegraded PE-MPs on *Brassica rapa* for 30 days and showed decreased plant growth (root and shoot length; fresh and dry weight). MPs treatment induces OS by releasing 2,4-epibrassinolide (DBP) that increases CAT and MDA levels. SEM analysis was also done which showed PE particles got attached and blocked the leaf stomata and lost their ability to open and close. The mechanism of MP toxicity is explored using transcriptomics in which MPs downregulated the genes related to ribosome metabolic pathway which are responsible for cell growth and proliferation. Also, there was an upregulation of plant-pathogen interaction pathway related genes which trigger Ca^{+2} dependent signaling pathway thus improves resistance of plants to external stress. A study by (B. Liu et al., 2024a, 2024b, 2024c) revealed the defense mechanism of plants against MPs. PS-MPs activate the phenylpropane metabolic pathway of cucumber with accumulation of metabolites in the leaves. These increased flavonoid metabolites (naringenin, pinobanksin) provided resistance to MP stress. Moreover, size dependent effect was seen in hydroponic condition where smaller PS-MPs showed strong translocation ability while larger PS-MPs showed stronger response (Hua et al., 2024; Zhou et al., 2023a, 2023b). Small sized MPs are major contributors to toxicity. However, effect of types, and different shapes of MPs on plants is limited (Hasan & Jho, 2023). More data based on diverse plant species is required, and it is also very important to comprehend how different MP types, and shapes affect plants. On the basis of the above- mentioned studies; various soil properties (physical, chemical and microbiological) get impacted by the MPs presence. Plant performance is indirectly affected by inhibition of root, shoot growth, reduction in yield due to seed germination Fig. 4.

2.2 Microplastic Impact on Aquatic Plants

MPs are toxic to aquatic plants (phytoplankton) which are primary producers of the aquatic system and crucial anchors for the stability of the food web. In order to adapt to pollution, phytoplanktons actively change the chemical makeup of EPS. It was discovered that the release of EPS rich in proteins aided in the formation of aggregates and surface modifications, which in turn affected the destination and colonization (Shiu et al., 2020). A batch culture of four diatoms was taken on which toxicology studies were already conducted using oil and related pollutants. Three sizes of PS-MPs 6 µm, 1 µm and 55 nm were used. They measured growth/survivial using extracted DNA and plotted it against protein/carbohydrate ratio (P/C ratio). The results revealed that nano sized plastics at low concentration led to lower P/C ratio than at high concentration. This further sheds light on the fact that high concentration of nanoplastic reduced survival of diatoms much more. To explore the combined toxicity of MPs and lead (Pb^{2+}) on submerged macrophytes (Ogo et al., 2022) conducted a 5 day hydroponic study. This showed decrease in physio-biochemical parameters such as soluble sugar, protein and chlorophyll content with increase in Pb concentration.

Thus, this study provides synergistic and species-specific toxicity. Similarly, (Z. Zhang et al., 2023a, 2023b, 2023c, 2023d, 2023e) investigated the effects of LDPE-MPs (0.03%, 0.3%, and 0.6%) of two different sizes. MPs decreased the biomass and abundance of rosette-forming species (*Vallisneria natans*) while increased that of canopy-forming species (*Hydrilla verticillata*). Thus, showed species-specific responses and effect on community structure where alpha diversity decreased at 0.6% concentration while enzyme activity related to nitrogen and phosphorus is increased. A multigenerational study was conducted for 10 generations on aquatic duckweed plant (*Lemna minor*) to investigate the effects of different sized PET-MPs. It revealed that small sized MPs negatively affected the photosynthesis only on acute exposure (3 day) while large sized MPs disrupted the chloroplast distribution and inhibited both the growth and photosynthetic activity on constant exposure. Few studies have been conducted so far on how MPs affect aquatic vegetation (R. Cui et al., 2023a, 2023b). After a long-term exposure of MPs, the aquatic macrophytes showed no significant changes on growth and biochemical properties revealing their resilience to MP stress. Upon seven days exposure, MPs adhered to plant biomass. About 25% of all MPs that were attached to plant biomass were strongly adhered. The mechanism behind the links between MPs and plant tissue is not fully understood, it is believed that the first interactions between MPs and plant biomass were electrostatic. Thus, these floating plants can be a used as a potential phytoremediation to remove MPs (Rozman et al., 2022). The aquatic plant growth is getting adversely affected depending upon the duration of exposure and the size of MPs. Also, based on the mentioned studies it is evident that MPs may exert plant species-specific toxicity.

3 Toxic Effect of Microplastics on Animal Health

Exposure to MP in animals including aquatic and terrestrial animal species has a number of detrimental health effects. For instance, MPs induces physical injury, inflammatory reactions, neural and even behavioral abnormalities and so on as discussed in subsections below.

3.1 MP Toxicity in Terrestrial Animals

MPs are also seen as a growing hazard to terrestrial ecosystems apart from aquatic ecosystem. Recent

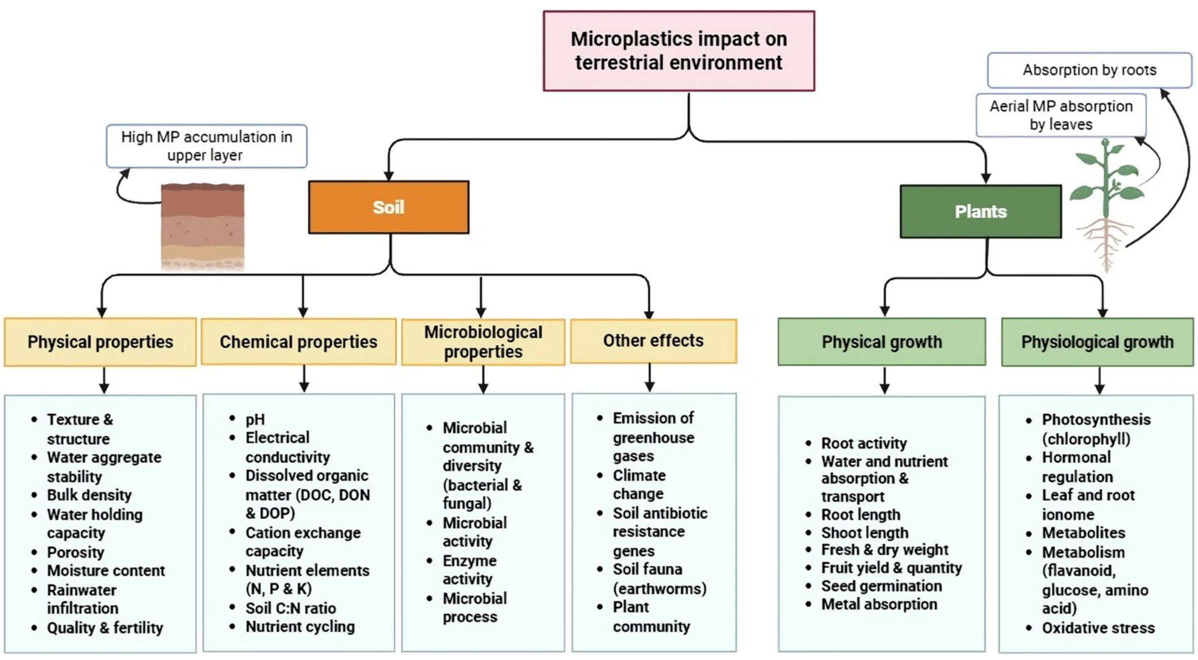


Fig. 4 Microplastic toxicity of terrestrial ecosystem including soil and plant ecosystem

research has revealed that terrestrial organisms are impacted by MP exposure through ingestion. MPs present in soil ecosystem adversely affect the fitness of organisms beneficial to agriculture and ecosystem. Studies were reported on earthworm (*Lumbricus terrestris*) to investigate the impact of MPs at 2.5%, 5% and 7% w/w concentrations. Avoidance behavior bioassay show that the worms were unable to discriminate between soils that contained MPs and ones that did not. MP adhered to their body producing physical lesions on mucus membrane when they move. The ingestion bioassay results showed that every earthworm segment contained MP at a specific concentration i.e. foregut at 7%w/w, midgut at 5%w/w, and hindgut at 2.5%w/w. These earthworm damages were measured using stereomicroscope with blue filter (Baeza et al., 2020). MPs and phenanthrene (Phe) co-exposure was found to be more genotoxic to earthworms; the concentration of MPs (10 μm) was found to cause DNA damage to earthworm coelomocytes (Xu et al., 2021). Earthworms were exposed to PE-MPs (two different sizes) for 21 days and showed minimal impact on the female reproductive organs but they harm the male reproductive organs. Hematoxylin and eosin (H & E) assays showed decrease number of mature sperm cells and increase in disordered seminal vesicle tissue in the exposed groups. Further, damage to the plasma membrane of sperm was confirmed by transmission electron microscope (TEM). The cytotoxicity assay revealed inhibition of cellular viability of coelomocytes by lowering intracellular esterase activity. Thus, MPs interfere with spermatogenesis of earthworm independent of size (Kwak & An, 2021). Beside this, PE-MPs even at environmentally relevant concentrations (0.01–0.5%), reduced the earthworm reproduction by 70%, resulting in a marked reduction in the production of juveniles exhibiting transgenerational effects (Sobhani et al., 2022). (Lahive et al., 2019) reported that small sized (13–18 μm) nylon MPs were found to be ingested more compared to 90–150 μm sized MPs as confirmed by the fluorescence microscopy. The impact on survival and reproduction of worm *Enchytraeus crypticus* either size dependence or dose dependence was measured. They found that the survival was unaffected but reproduction was significantly affected at higher MP doses. Terrestrial snails (*Achatina fulica*), i.e. another widely distributed soil invertebrates were exposed to PET fibers for 28 days. The results demonstrated that after intake of MPs, cracks on the surface of MPs post depuration were observed while passing through digestive tract. Prolonged exposure caused significant damage to 40% snails via damaging the villi of gastrointestinal walls but did not affect the histology of kidney and liver. It suppressed the enzymatic activity of glutathione peroxidase (GPx) and decreased T-AOC levels and elevated MDA levels thereby inducing toxicity through OS (Song et al., 2019). (de Souza et al., 2022) reported the first study linking the consumption of MPs by small terrestrial birds to biochemical changes, such as OS prediction, and cholinesterase effect. A large amount of MPs were released in feces, which can act as a potential carrier of dispersal of MPs in the environment. Thus, the soil invertebrates inhabiting MP in MP contaminated soil surface faced adverse effects on reproductive health via ingestion of smaller MPs.

3.2 MPs Toxicity in Aquatic Animals

It has been reported that more than 150 fish species and aquatic organisms such as zooplanktons, fishes, mussels, crabs, and shellfishes have been discovered to have accumulated significant amounts of MPs (W. Wang et al., 2020a, 2020b). The unintentional ingestion of MPs occurs mainly due to the direct uptake of MP contaminated food rather than through aquatic food chain transfer because of their diminutive size. Two possible mechanisms exist for the uptake of MPs: phagocytosis and pinocytosis. One of the findings strongly imply that animals can absorb MPs through adherence (50%) rather than ingestion. All organs of mussels (*Mytilus galloprovincialis*), including the foot and mantle, showed signs of microfiber accumulation and retention (Kolandhasamy et al., 2018). When disposed; the MPs reach aquatic sources. And those of particular size and concentration get ingested or adsorbed on their body surface and have consequent adverse effects on the aquatic organisms such as neurotoxicity, inflammation, reproductive and developmental toxicity and so on as shown in Fig. 5. There is still a lot to learn about the impact of the intricate combinations and potential interaction of pollutants with MPs and their impact on the organisms. MP-organism interactions enhance their bioaccumulation and toxicity (Xu et al., 2020). The epibionts living on plastic surface have birthed a new era of plastisphere.

3.2.1 Neurotoxicity and Behavioral Alteration

MP exposure led to neurotoxicity which can be induced via reactive oxygen species (ROS) mediated pathway or nitric oxide (NO) mediated pathway. Acetylcholinesterase (AChE) is used as a potential biomarker and an indicator of neurotoxicity. AChE enzymes are necessary for cholinergic neurotransmission in cholinergic brain synapses and neuromuscular junctions. Due to MP toxicity, acetylcholine (ACh) levels significantly rise and accumulate in the synaptic cleft as a result of the suppression of AChE activity which leads to neuromuscular and neurological dysfunction (Wen et al., 2018a, 2018b). Further, (Kim et al., 2021) reported that AChE inhibition destroys lipid peroxidation (LPO) and upsets the nerve related enzymes in the brain of the fishes. Similar results of AChE inhibition due different MP exposure were reported in various fishes such as African catfish (*Clarias gariepinus*) (Iheanacho & Odo, 2020), wedge clam (*Donax trunculus*) (Tlili et al., 2020), crucian carp (*Carassius* Carassius) (Barboza et al., 2018; Wen et al., 2018a, 2018b). (Barboza et al., 2018; Wen et al., 2018a, 2018b) reported behavioral alteration in swimming performance by observing reduction in resistance time, swimming velocity and lethargic swimming of European seabass (*Dicentrarchus labrax*) following a 96 h of exposure to MPs or to Mercury alone and also in concoction. Limited studies are available that show the combined impact of MP with other pollutants like heavy metals. (Santos et al., 2021) investigated effect of MPs (2 mg/L) along with copper (Cu, 60 & 125 µg/L) on zebrafish larvae 14 days post fertilization (dpf). The findings suggested possible deficiencies in the neurogenesis of the exposed zebrafish at early life stages by demonstrating a downregulation of genes linked to neuronal proliferation viz (*sox2*, *pcna*), neurogenesis (neuroD, olig2), and motor neuron development (islet) thereby impairing the cellular proliferation in brain. AChE can be used as potential biomarker for assessing MP and copper toxicity. (Santos et al., 2022) further conducted a prolonged exposure of 30 days to find behavioral changes and altered antioxidant system in brain. This was estimated by high of levels superoxide dismutase (SOD), glutathione reductase (GR) indicating OS in fish due to increased AChE. A rise in the expression of apoptosis-related genes (*caspase3*, *caspase8* and *caspase9*) was noted. MP induced apoptosis by the depletion of GPx which has antiapoptotic property and inhibits the release of cytochrome c from the mitochondria thus, activating caspase-3. Another mechanism related to altered dopaminergic pathway which increases dopamine (DA) synthesis by the upregulation of tyrosine hydroxylase (*th*) and dopamine transporter (*slc6a3/dat*) may also be involved. DA enters into cells and oxidizes to form ROS; thus activating the caspase pathway. Further (Suman et al., 2023) underlined the molecular mechanism for neurotoxicity by investigating health of zebrafish embryos against the PS-MPs. 500 nm PS-MPs get ingested and accumulated in the gut region in a dose dependent manner. But MPs had no detrimental effects on the embryos'survival or rate of hatching, nor did they induce any lethality. PS-MPs reduced the SOD and CAT activity and which resulted in OS. Acridine orange staining and gene expression using qRT-PCR were performed. MP lowers the AChE activity and in turn increases the NO concentration along with altered serotonin (5-HT), DA and downregulation of bdnf genes. This further induces downstream apoptotic signaling pathway by upregulating p53 and downregulating Bcl-2 gene expression. *caspase3* and *caspase9* expression is upregulated thereby inducing neurotoxicity. A recent study by (X. Li et al., 2024a, 2024b, 2024c) revealed that aged PS-MPs pose greater neurotoxicity compared to virgin PS-MPs. They inhibited the motor neuron development via disrupting hb9-GFP receptor and thus altered AChE enzyme activity and neurotransmitter release causing neuronal damage and aberrant neurotransmission. Aged MPs at a dose (0.1 to 100 µg/L) also altered the expression of genes *5ht1aa*, *dat*, and *gabral* related to the neurotransmitters that increased neurotransmitter levels (DA, 5-HT, and GABA) besides showing a negative effect on swimming behavior of zebrafish (Xiang et al., 2023). Additionally, MPs with other environmental pollutants like triphenyltin (TPT) impair locomotory performance by affecting the gene related to neural development and thus exacerbate neurotoxicity (Lin et al., 2024). Similarly, co-exposure of clothianidin (CLO) and aged MPs enhances neurotoxicity through neurotransmission disruption i.e. significant changes in the neurotransmitter levels such as ACh, DA, GABA, 5-HT were

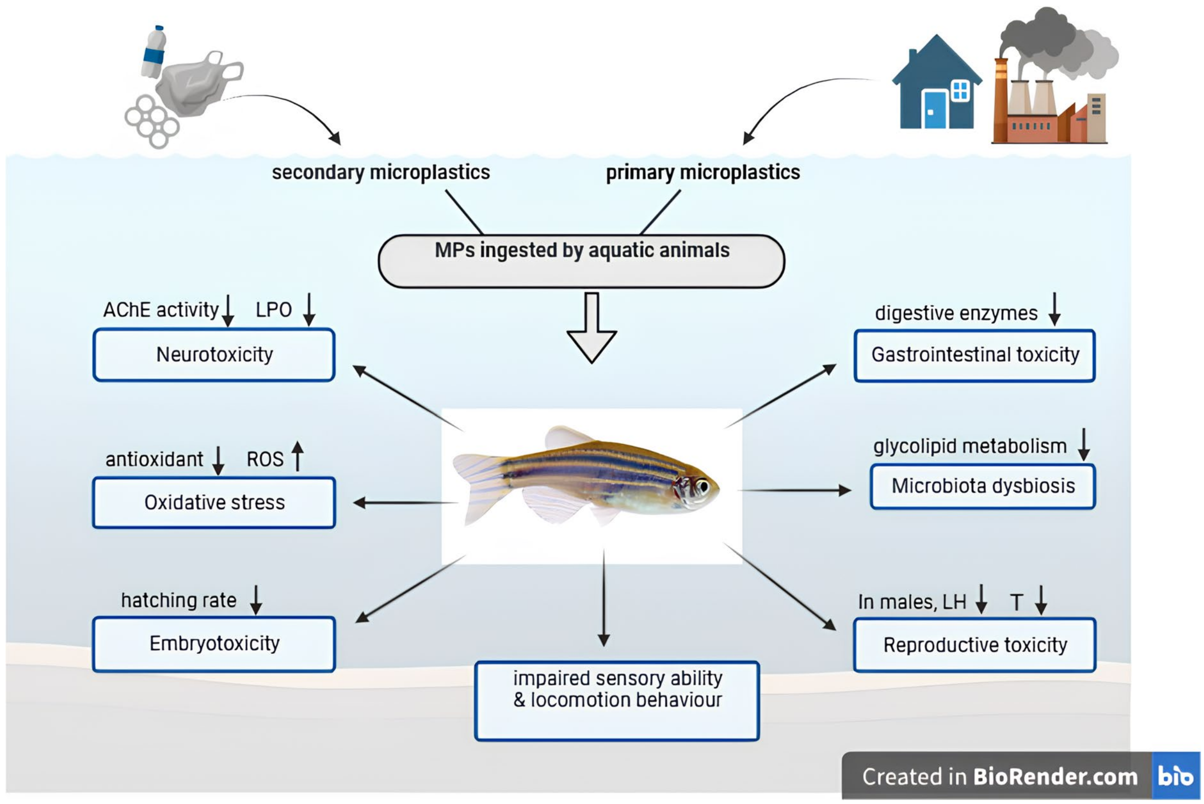


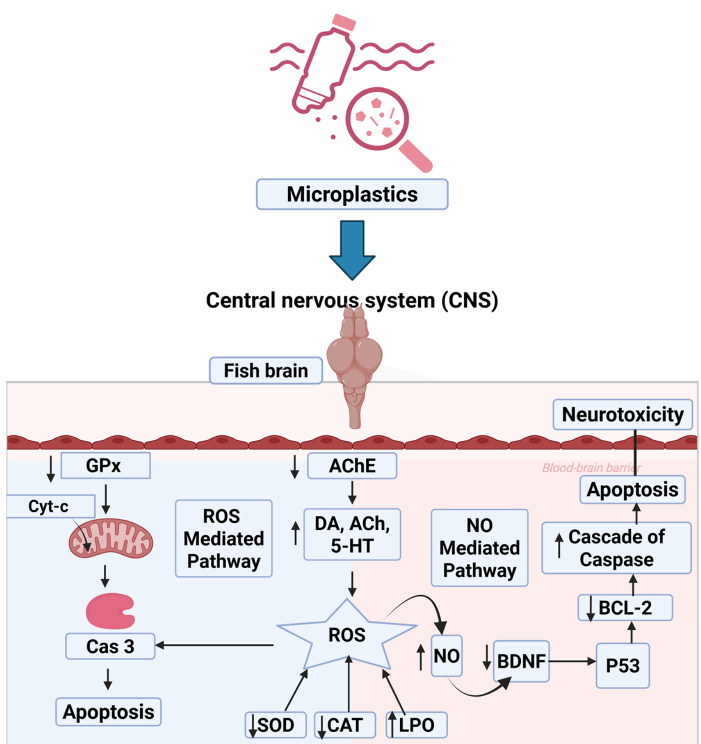
Fig. 5 Diagram showing toxic impact of MPs on aquatic animals

observed. This reduced locomotory activity. Hypothalamic-pituitary-interrenal (HPI) axis analysis showed increased levels of adrenocorticotropic hormone (ACTH) and cortisol, along with downregulated expression of stress-related genes which indicate impairment of neuroendocrine system (Ding et al., 2025). Thus, MPs impact the health of fish by promoting OS, disturbing LPO and antioxidant system which finally causes neurotoxicity in a dose and time dependent manner. The mechanism behind this is illustrated in Fig. 6.

3.2.2 Inflammation and ROS Responses

MPs accumulation in fish generate excess ROS and triggers antioxidant responses. The imbalance between ROS and antioxidant capacity induces OS. Also, MDA that is an end product of LPO is indicative of the OS. OS and inflammatory reactions are important mechanisms that underlie damage caused by MPs. (Lu et al., 2018) revealed that ROS was produced as a result of adult Zebrafish's chronic exposure to a combination of PS-MPs (20–200 g/L) and Cadmium, Cd (10 g/L), significantly reducing the activity of glutathione (GSH), SOD and thus causing OS. Also, upregulated expression of *tnfa*, *il1b* and *ifng1-2* are responsible for inflammatory responses in tissues. MPs enhanced the buildup of Cd in zebrafish tissues mainly in liver when exposed for 3 weeks. But Cd accumulation was found to be in order gill > gut > liver. This provides evidence for the chronic effects of MPs along with heavy metals. An accumulation of MP in gut of sheepshead minnows (*Cyprinodon variegatus*) were observed upon exposure to 50 and 250 mg/L of PE-MPs (irregular and spherical). This caused intestinal distention, generated intracellular ROS and induced OS by upregulating genes (*Sod3* & *Cat*) as observed by SYBR-Green-based quantitative real time PCR. Additionally, the results showed irregular MPs interaction at molecular level led to an increase in transcriptional levels of *Tp53* and *casp3* suggesting that apoptotic processes can be influenced by MPs (Choi et al., 2018). Thus, it showed toxicological impact of environmentally relevant microparticles on marine fishes. In a study by (Capó et al., 2021) Gilthead seabreams (Sparus aurata Linnaeus, 1758) fish were fed with LDPE-MPs (100 and 500 μM) for 90 days followed by 30 days depuration. They found that fish growth was unaffected throughout the study but an alteration of antioxidant activities (higher CAT and GR) was observed. Increased MDA contents which is an indication for OS were observed. An increased myeloperoxidase (MPO) activity which showed inflammatory responses in liver was also noted. After depuration, the oxidative damage markers can recover to its initial values but not completely. The same fish fed a diet enriched with PS-MPs (0, 25, or 250 mg/kg) for 21 days and had no effect on the physiological growth but inflammation of the anterior and posterior intestine was observed. Ingested PS-MPs alter the cytokines release pattern via activation of TLR/MyD88 signaling pathway, which further activates p38 and ERK via the mitogen-activated protein kinases (MAPK) pathway. This disrupts posterior intestinal epithelial integrity by reducing the expression of tight junctions (*ZO-1*, *Cldn15*, Occludin), and mucin (*Muc2-like* and *Muc13-like*) (Del Piano et al., 2023). Further, a study by (J. Cui et al., 2023a, 2023b) conducted on carp fed with PS-MPs (1000 ng/L) along with water to investigate the toxicological mechanism leading to hepatopancreatic inflammation was conducted. A hepatopancreas function test showed abnormal increases in alkaline phosphatase (AKP), alanine aminotransferase (ALT), aspartate aminotransferase (AST), and lactate dehydrogenase (LDH) activities which led to tissue damage and aberrant hepatopancreatic activity as confirmed by H&E staining. An inhibition of antioxidant enzymes and accumulation of ROS induced OS was noted. ROS activates TLR2/MyD88/TRAF6/NF-κBp65 pathway and increases the release of pro-inflammatory molecules such as tumor necrosis factor (TNF-α), interleukin-1β (IL-1β), inducible nitric oxide synthase (iNOS), and cyclooxygenase 2 (COX2) which were analyzed through qPCR, western blotting and enzyme-linked immunosorbent assay (ELISA) in hepatopancreas and hepatocytes. Furthermore, PE-MPs induce OS by NF-κB pathway. This further regulates nucleotide-binding oligomerization domain-like receptor protein 3 (NLRP3) inflammasome expression detected by the immunofluorescence staining. NLRP3 increases expression of immune inflammatory factors

Fig. 6 Mechanistic pathway of MP inducing neurotoxicity in aquatic animals



resulting in apoptosis and inflammation in the gills of the carp as analyzed by TdT-mediated dUTP nick-end labeling (TUNEL) assay and RT-PCR (Cao et al., 2023a, 2023b). Also, (Xu et al., 2023) revealed the molecular processes behind MP-induced inflammation and apoptosis including ferroptosis induction via activation of the NF-κB signaling pathway. PS-MPs also damage the myocardial tissue of the carp (Q. Zhang et al., 2023a, 2023b, 2023c, 2023d, 2023e) provide the specific mechanism behind this. This is due to the activation of TLR4/MyD88/TRAF6 pathway followed by NF-κB pathway. The inflammatory factors get released in a dose dependent manner. A recent study by (T.-j. Liu et al., 2024a, 2024b, 2024c) elucidated the mechanism of PE-MPs (1000 ng/L) on the muscle tissues of the carp fed for 21 days. This muscle injury is caused by the ROS generation and by the OS generated. The upregulation of microRNA (miR-21)/Interleukin-1 Receptor Associated Kinase 4 (IRAK4)/NF-κB pathway, lead to inflammation. In particular, NF-κB pathway acts as an important mediator connecting OS and inflammation. Thus, sub-chronic exposure of microparticles induces inflammation in freshwater and marine fishes.

3.2.3 Biochemical and Hematological Changes

MPs in water diffuse through cell membrane into the bloodstream of fish and accumulate in major organs through blood. Also, MPs with ≤5 μm can directly enter the circulatory system through gills and impact the hematological parameters (Jovanović et al., 2018; Kim et al., 2021). Hematological parameters are crucial biomarkers for assessing the toxic physiology of fish as well as of their health status on MP exposure. (Hamed et al., 2019) reported that upon exposure to MPs at 1, 10, and 100 mg/L for 15 days, some of the biochemical parameters like AST and ALT activity as well as uric acid levels, glucose and creatinine were higher in Nile tilapia (*Oreochromis niloticus*). The high levels in bloodstream of fishes suggested that the MPs cause serious damage to the cell membrane and also deteriorated the renal and hepatic tissues. While reduced hematological parameters (red blood cells (RBCs), hemoglobin (Hb), platelets, hematocrit (Ht), and white blood cells (WBCs) count) were observed which resulted in hemolysis and anemia ultimately causing the mortality of juvenile fishes. An investigation carried out on common freshwater carp (*Cyprinus carpio*) confirmed the presence of PE-MPs and its accumulation in liver and intestine. A perturbation in liver biochemical homeostasis was observed upon exposure of 30 days. Total protein, butyrylcholinesterase (BChE) and globulin levels declined while the biochemical parameters such as alkaline phosphatase (ALP), creatine phosphokinase (CPK), (LDH), gamma-glutamyl transferase (GGT), AST, and ALT increased in MP treated fishes. Also, fish exposed to MPs had noticeably greater levels of glucose, creatinine, cholesterol, and triglycerides (Banaei et al., 2022). Further, (Yu et al., 2023) also showed concentration dependent effect irrespective of the exposure duration on the hematological parameters and plasma components. Moreover, variations in temperature (17 and 22 °C) also impacted the biochemical parameters, and hematology of Crayfish (*Astacus leptodactylus*) due to PE-MP exposure (Gholamhosseini et al., 2023). (Raza et al., 2023) explored the occurrence of polyacrylamide (PAM) MPs in the gills of tilapia fish. High doses of MPs resulted in decreased antioxidant enzymes, Hb levels and other blood parameters. Also, an increase in MDA and total protein contents was noted. Histopathological alterations in gills, intestine and liver of the fish were also seen. Thus, higher doses of PAM-MPs affected the fish health. The toxicity of polyamide (PA) MPs (5000 and 10,000 mg/L) through consumption and accumulation were tested for 96 h in juvenile Korean bullhead (*Pseudobagrus fulvidracoi*) and an accumulation profile was found to be gut > gill > liver. This study reported a dose dependent effect with decreased hematological parameters while remarkably increased biochemical parameters. Additionally, the plasma components including magnesium, total protein and calcium were reduced and the antioxidant response of the fishes was affected (Jo et al., 2025). In India (Kerala), an investigation was carried out on green chromide (*Etroplus suratensis*), a state fish of Kerala which showed toxicological effects due to PVC-MPs exposure. They found altered antioxidant enzyme activity, and reduced blood parameters along with the behavioral changes (Vijayaraghavan et al., 2022). The survival and wellbeing of the endangered turtle species could be at risk due to MP exposure. Therefore, pond turtles (*Emys orbicularis*) were studied for investigating negative impact of MP through dietary exposure of PE-MPs (250, 500, and 1000 mg/kg) for 30 days. Blood biochemical parameters were impacted and reduced amount of total immunoglobulins, albumin, phosphorus, total protein and also triglyceride levels were reported which resulted in tissue damage (Banaee et al., 2021). Additionally, combined toxicity of PE-MPs and 4-nonylphenol (4-NP), a chemical pollutant was investigated. The synergistic effect decreased hematological parameters and majority of hematobiochemical parameters improved during the 15-day recovery period (Ammar et al., 2023). Therefore, prolonged exposure to MPs could cause significant changes in the blood parameters and majority of toxicological research is conducted at the individual and tissue levels; more cellular and genetic studies are required to fully comprehend MP toxicity.

3.2.4 Embryonal Changes and Developmental Toxicity

MPs gets easily adsorbed onto the embryonic chorion due to its negative zeta potential. It changes the mechanical properties of the chorion by blocking and affecting the permeability of chorion channels thereby reducing oxygen delivery which leads to hypoxic environment in the embryos. The consequent developmental toxicity changes are in heart rate, blood flow velocity and hatching rate (Cheng et al., 2021; Duan et al., 2020). (Y. Li et al., 2020a, 2020b) reported the pooled toxicity of PS-MPs (0, 2, 20 & 200 µg/L) and phenanthrene (Phe) (50 µg/L) on fertilized eggs of marine medaka (*Oryzias melastigma*) through consumption by the larvae. This study demonstrated that over the course of 5–10 days post exposure, MPs with 10 µm diameter attained a plateau on the egg shells of medaka due to adsorption on the sticky chorion reaching a saturation point. High levels of MPs decreased hatching, and growth but did not alter Phe toxicity. Co-exposure to 2 µg/L MPs and Phe, elevated the hatchability by 25.8%, malformation and mortality rates dropped, and Phe-induced aberrant expressions of genes linked to heart development were restored. Furthermore, a study by (Xia et al., 2022) revealed the toxic effect of PVC secondary MPs (SMP) due to their irregular shape and slight aggregation causing physical damage to the chorion surface as compared to the primary MPs (PMP). This damage causes changes in heart rate, malformation and altered hatching time in marine medaka. (De Marco et al., 2022) tested zebrafish for embryotoxicity upon treatment with PS-MPs (10 µm) for 120 h post fertilization (hpf) and found significant abnormalities in larval development, primarily at the level of the tail and column, as well as weakened integrity of eyes. But, in developing zebrafish; the survival was not impacted, only a moderate delay in hatching was found. In India, KRS-CR (Krishnaraja Sagara dam-Cauvery River) water with an abundance of MPs was used on zebrafish embryos to test embryotoxic effects. Elevated production of ROS, which causes subcellular organelle dysfunction, skeletal deformities, DNA damage and increased mortality were observed (Anifowoshe et al., 2022). Due to discharge of PE microbeads (facial cleansers) into aquatic bodies; an adverse effect on several zebrafish embryos was reported i.e. early hatching, increased heart rate, increased teratogenic malformation at high doses of MPs (De Guzman et al., 2020). A study demonstrated that in sea cucumber (*Apostichopus japonicus*) embryo-larval development is affected by both PVC-MPs and its leachates with leachates exhibiting greater toxicity (H. Wang et al., 2023a, 2023b, 2023c). In addition, MPs with other contaminants such as di(2-ethylhexyl) phthalate (DEHP) (H. Wang et al., 2022a, 2022b, 2022c), sulfamethoxazole (SMZ) (Lu et al., 2022), and 3,6-dibromocarbazole (3,6-DBCZ) (J. Zhang et al., 2024a, 2024b, 2024c) had an antagonistic effect and also activated NF-E2-related factor 2 (Nrf2) signaling pathway to reduce OS toxicity in larval zebrafish. In another study, zebrafish larvae were exposed to PE-MPs (20–27 µm) with benzo[α]pyrene (BaP) (1%) supplemented with their food. They evidenced adverse effects after 30 dpf including increased skeletal deformities with impaired caudal fin/scale development. When parents were exposed to MP-BaP; an impairment in bone formation in the offspring was observed particularly in the opercular bone in 6 dpf larvae throughout generations (Tarasco et al., 2022). Co-exposure of MPs and 2,2′,4,4′-tetrabromodiphenyl ether (BDE-47) revealed detrimental developmental toxicity in *Hexagrammos otakii* including decreased heart rate, higher mortality, body deformity, and decreased hatchability by activating Wnt signaling pathway (Shi et al., 2024). One of the recent findings challenges the earlier theories regarding barrier function of chorion against larger MPs demonstrating that PE-MPs 1–4 µm can penetrate the natural barrier like chorion and reduced the heart rate by downregulating the heart development genes (*fbln1* and *fn1b*) (Y. Kim et al., 2025a, 2025b). Therefore, co-exposure of MPs with other contaminants can show synergistic and antagonistic affect that induces developmental toxicity and there is a need to explore the mechanism of development toxicity which is still lacking.

3.2.5 Gastro-Intestinal Alteration

MPs aggregate in GI tract, possibly causing aberrant behavior in fish. Continuous consumption of MPs may result in intestinal lumen distention, which could ultimately result in intestinal obstruction (Yin et al., 2018). A study was conducted to investigate size dependent effects of HDPE-MPs (< 100 µm) on zebrafish larvae wherein gut damage by altering the morphology of GI tract was reported (Kim et al., 2022). (Zhao et al., 2021) also explored the length dependent intestinal toxicity of MP fibers on the zebrafish. Long MP fibers significantly reduced the zebrafish's dietary intake to 54% and 67% at 3 and 10 min, respectively as compared to short MP fibers. Potential explanations include increased glycerophospholipid metabolism, worsened inflammation and oxidative damage, and decreased control of the metabolism of fatty acids associated with malnutrition. Furthermore, it was found that MPs containing PLA (bio-based plastics) were actively consumed and accumulated in intestines 170 times higher than the PET, (petroleum-based plastics) in zebrafishes and resulted in gastrointestinal damage and alteration in diversity of intestinal microbiota (Duan et al., 2022). It has been reported that zebrafish exposure to PUR-MPs for 10 days, causes tissue damage thus, significantly increasing fish mortality. It showed toxicity primarily in gills, GI tract and liver. Congestive inflammation, epithelial damage, bowel wall weakening, lesions of the villi and epithelial separation in the gastric wall were among the observed GI damages (Sadat Hosseini Dinani et al., 2021). Likewise, the toxicity of PS-MPs (100 and 1000 µg/L) on guppy (*Poecilia reticulata*) was tested for effects on gut upon 28 days exposure. An enlargement of goblet cells and reduced activity of digestive enzymes (trypsin, amylase, lipase and chymotrypsin) along with a decreased diversity of gut microbiota was observed. This resulted in disturbed digestion and gut dysbiosis (Huang et al., 2020). (Lei et al., 2018) showed effects of MPs (PE, PA, PVC and PP) having size ~70 µm at various concentrations ranging from 0.0001 to 10 mg/L on zebrafish after 10 days of exposure. They experienced intestinal damage primarily manifested as villi cracking and enterocyte splitting. A size-dependent negative impact PA-MPs on survivability and intestinal abnormalities of juvenile striped catfish (*Pangasianodon hypophthalmus*) was revealed. Smaller MP (25–50 µm) accumulated more in the GI (Shahriar et al., 2024). Hence, size and length dependent long-term exposure to MPs may have an impact on fish nutrition, which may then influence fish health and growth.

3.2.6 Microbiota Dysbiosis and Metabolism Alteration

Studies show the noxious effects of MPs on alteration of microbial community structure of aquatic organisms. MPs were found to accumulate and exert toxicity in the shape dependent manner with fibers > fragments > beads. It causes intestinal toxicity such as inflammation, increased permeability, and gut dysbiosis in the zebrafish (Qiao et al., 2019). In another study, PS and HDPE accumulated in tissues of mussels in order: digestive gland > gill > gonad ≈ muscle. MPs induced changes in the metabolic profile and majority of the PS and HDPE were eliminated by mussels after 144 h (Wei et al., 2021). Both size and time dependent effects of MPs were reported on zebrafish gut microbiota wherein difference in gut microbiota composition upon different treatments was observed. This study showed that the abundance of *Bacteroidetes* and *Actinobacteria* significantly increased while the relative abundance of *Tenericutes*, *Firmicutes* and *Verrucomicrobia* significantly decreased in MP exposed groups (Xue et al., 2021). The fate of PE fibers at 1% concentration consumed by *Lates calcarifer* and its impact on intestinal health were examined by (Xie et al., 2021). Low fiber retention was noted, suggesting that fibers were successfully removed from the fish's body. Fibers (2–3 mm long) influenced the growth of the fishes but long-term exposure led to chronic effects resulting in intestinal damage due to increased MDA levels. Microbiota dysbiosis as observed by reduced alpha diversity, particularly community richness, and inhibition of proliferation of *Lactobacillus* were also observed. PE particles were discovered to accumulate in the crayfish's intestine, hepatopancreas, gills, and hemolymph following a 21-day food exposure. And MPs were still retained in these tissues after 7 days depuration period in clean water. It impacted the microbiota balance by decreasing the overall microbiota abundance and changing the ratios of numerous bacterial families (X. Zhang et al., 2022a, 2022b, 2022c, 2022d, 2022e, 2022f). The effect of PS-MPs of various sizes was investigated on adult marine medaka and it was found that larger MPs induce gut microbiota dysbiosis and also altered lipid metabolism in liver. They do so by disturbing the composition and diversity of microbes in the gut of fishes. Overall, the findings proved that MPs exposed fishes have abundant bacterial taxa mainly belonging to the phylum *Firmicutes*, *Fusobacteria* and *Verrucomicrobi*a (Zhang et al., 2021). The gut microbiota of carp was disturbed upon exposure of PS which also induces inflammation, endoplasmic reticulum stress, and apoptosis in the intestinal tissues. There was a significant decrease in *Bacteroides*, *Pauro-saccharolyticus*, *Lactococcus garvieae* and *Romboutsia ilealis* in intestinal flora (F. Wang et al., 2023a, 2023b, 2023c). European sea bass (*Dicentrarchus labrax*) juveniles were fed on PP-MPs in combination with chemical pollutants for 60 days. The changes in the gut microbiota composition and bacterial species richness revealed raised levels of potentially harmful microorganisms (*Proteobacteria* and *Vibrionales*). A scaling down of helpful lactic acid bacteria that caused intestinal dysbiosis along with inflammatory like response in distal intestine (Montero et al., 2022) were observed. Furthermore, a combined exposure of MPs (1 µm) and bisphenol-A (BPA) 100 µg/L potentially damaged the intestinal villi and disturbed the intestinal microbial community of crab (X. Chen et al., 2024a, 2024b, 2024c, 2024d). This gave a glimpse that MPs can modify the composition of gut microbiota in shape, time and size dependent manner.

3.2.7 Immune Response

MPs impact the innate immune function of fishes. For instance, hybrid snakehead (*Channa maculate* × *Channa argus*), an economically important fish was also tested for MP toxicity at two stages of development. It was reported that the larvae were more sensitive to high concentrations of small MPs and, showed high mortality rate. Juveniles showed stimulated immune response and also intestinal morphology damage (C. Zhang et al., 2022a, 2022b, 2022c, 2022d, 2022e, 2022f). A study on edible mussel tested for MP toxicity (PE microbeads) in marine ecosystem showed disrupted homeostasis upon 18 days of exposure with generation of immune and stress associated proteins. And, a consequent reduction in the energy content was noted. In the recovery period, the expression of these proteins continued. After second exposure, a reduction in protein expression was noted in digestive gland suggesting that a different response developed in mussels (Détrée & Gallardo-Escárate, 2018). An inhibitory impact of MPs at higher concentration (40 mg/L) or extended exposure period was observed on innate immune system with regards to intestinal microbiota of juvenile Chinese mitten crab (*Eriocheir sinensis*). Hepatopancreas exhibited greater negative response compared to the hemolymph or hemocytes. The amount of immune related factors decreased in hepatopancreas while the hemocyte expression of myeloid differentiation factor 88 (*MyD88*) and *caspase* was higher due to MP-induced stress (Liu et al., 2019). Similarly, white leg shrimp (*Litopenaeus vannamei*) was subjected to various concentrations of MPs for 48 h to study their immune response and microbiota. Shrimps had high immune and microbial response and low survival rate when exposed to high doses of MPs (5000 µg/L) (Z. Wang et al., 2021a, 2021b). PE-MPs activated the intestinal immune network in a dose dependent manner by altering the innate immunity complement C3 and C4. It can potentially increase the inflammation probability in the intestinal mucosa by altering the intestinal microbiota (Yuan et al., 2023). The toxic impact of PS-MP (5 µm) was explored on immune response of fishes including damage to kidney by lipid accumulation. PS-MPs inhibited the innate immune pathway i.e. cytosolic DNA-sensing pathway which causes downregulation of the immune factor interleukin-1 beta (*il1b*) thus increased the susceptibility to infections (Yang et al., 2024). MPs as foreign substances have the potential to either stimulate fish immune responses or depress immune function by causing immunological toxicity. Specifically, the innate immune system along with the intestinal microbiota of the fishes was affected at higher concentration of MPs.

3.2.8 Reproductive Damage

MPs cause an adverse effect on the reproductive function of the fishes. The hypothalamic-pituitary–gonadal axis (HPG axis) is a key regulator of fish reproduction. The pituitary produces and secretes gonadotropic hormones (GtH, including FSH and LH) in response to GnRHs released by the hypothalamus. GtH then initiates steroidogenesis, gametogenesis, and gonad maturation which are responsible for sex hormone production. MP (2, 20, and 200 μg/L) exposure to marine medaka had negative effect on the HPG axis as confirmed by the transcriptional analysis. This show reduced transcriptional levels (*StAR*, *11βHSD*, *17βHSD*, *CYP11a2*, and *CYP17a1*), hence suppressing the steroidogenesis pathway. Gonad maturation was also postponed. Vtg and Chg (used for oogenesis) got inhibited which decreased 17*β*-estradiol (E2) level. This resulted fertility reduction in female fish of marine medaka. Additionally, when parents gets exposed to 20 μg/L MPs, it affected the offsprings too by decreasing the heart rate, body length, and hatching rate (Wang et al., 2019). (Ismail et al., 2021) examined the negative consequences on the male reproductive performance of the tilapia fish when fed with increasing concentration of supplements of (0%, 2.5%, 5%, and 7.5%) *Amphora coffeaeformis* (diatoms) for 70 days and then transferred to another aquarium containing MPs (10 mg/L) for another 15 days. They observed reduced luteinizing hormone (LH) and, testosterone (T). Testicular alteration (testis-ova) was also observed within the testicular lobules and, a sizable number of early perinucleolar oocytes developed in MP treated groups. But the results show that high levels of supplementation of *A. coffeaeformis* avert the negative impact caused by MPs as the testis morphology improved. This study suggests that *Amphora* feeding can be the remedial for plastic pollution. At a lower dosage of PS-MPs (10 μg/L), no effect was seen on reproductive organs of zebrafish. A continuous exposure of 21 days of high dose of MPs (100–1000 μg/L) increased ROS in liver and gonads of both female and male zebrafish. Histological changes were also seen such as a considerable scaling down in the thickness of the testis basement membrane. Male testicles revealed a notable increase in level of apoptosis, as observed by increased p53-mediated apoptotic pathway expression (Qiang & Cheng, 2021). Thus, ROS level in fish gonads can serve as a reliable early warning sign of impaired reproductive system. Furthermore, the findings indicate that chronic exposure of PVC-MPs (2 ± 1 μm) on *Daphnia magna* promoted the overall quantity of broods of each female and the frequency of molting per adult, bringing down the number of offspring at the first brood. The overall number of offspring produced by each female was also brought down (Y. Liu et al., 2022a, 2022b). Conversely, though 50 ± 10 μm sized MPs only declined offspring at first brood indicating that MPs caused size-dependent reproductive damage in them. Freshwater zooplankton, waterflea (*Ceriodaphnia dubia*) were also reported to have reproductive health problems when treated with PES fibers in acute and chronic phase. Upon MP exposure of 48 h, waterflea for showed dose dependent effect on survival while 8 days exposure adversely affected the reproduction as well as growth (Ziajahromi et al., 2017). A study in prawns reported that MPs (2 and 20 mg/L) induce OS in testis which inhibited male testicular development. Also, parental exposure to MPs reduced offspring larval immunity and survival rate (S. Sun et al., 2022a, 2022b). Adult loaches (*Paramisgurnus dabryanus*, F0 generation) were exposed to PE-MPs (1 and 10 mg/L) for 30 days and negative effects on the offspring (F1 generation) were assessed. This demonstrated that exposure to PE-MPs could alter the antioxidant system in the liver, gonad and brain of adult loaches. PE-MPs build up in the gonads and interfere with the transcription of genes related to the HPG axis which alters the sex hormone levels. Also, increased gonadal pathological lesions were observed along with degradation of biological properties of semen indicating reproductive toxicity. Accumulated PE-MPs got transferred from parental gonads to the artificially inseminated offsprings and got aggregated on the chorionic membrane of embryos. They raised malformation, mortality rates, and reduced the hatching rate and body length (Xia et al., 2023). Trans-generational toxicity of PS-MP (Rong et al., 2024), and PLA-MP (L. Zhang et al., 2024a, 2024b, 2024c) have also been reported. PS-MPs can have deleterious effect on HPG axis via sirtuin 1 (SIRT1) modulation that causes oocyte maturation failure and changes in plasma steroid hormone levels (E2/T ratio). These indicate hormonal imbalance and reproductive impairment in zebrafish. Also, as investigated through molecular docking, PS-MPs effectively bind and inhibit endocrine receptors and SIRT1 (Gupta et al., 2023). Combined toxicity of HDPE-MPs and DEHP revealed reproductive dysfunction in female mussels. Ovarian tissue's antioxidant capability was reduced, ovarian growth was hampered and reproductive function decreased due to the reduction of the gene expression of ER of cytochrome P450-3 (CYP3), and 17β-hydroxysteroid dehydrogenase (*17β-HSD*), which in turn decreased the levels of progesterone and estradiol (X. Lan et al., 2025a, 2025b). Hence, fish gonad exposed to MPs can pose a potential negative impact on reproductive health and also on the offspring health by transgenerational toxicity. These studies offer new insight on the cross-generational effects of the microplastics.

3.2.9 Other Behavioral Effects

MPs have been demonstrated to reduce the locomotory behavior of aquatic species by 6% at environmentally relevant concentration of ≤ 1 mg/L. This played a key role in feeding and social activities of aquatic organism (T. Sun et al., 2021a, 2021b). Findings proved that MPs impaired the sensory ability of goldfishes in response to odorants (Shi et al., 2021). Upon 28 days exposure to PS-MPs, the olfactory behavior of goldfishes changed due to suppression of olfactory related genes (G-protein coupled receptor) and inhibition of action potential generation via the ATPase. Also, the olfactory neural signal transduction was disrupted causing olfactory bulb injury. Large sized HDPE-MPs caused a destruction to mechanosensory receptors and hair cells in fish lateral line system (Kim et al., 2022).

Overall MPs have a detrimental effect on the health of the aquatic organisms including phytoplanktons and, fishes at molecular, cellular and individual levels. Table 1 shows effects MPs on organisms at varied concentration, size and types.

4 Toxicity and Adverse Impact of Microplastics on Human Health

Exposure Routes of MPs MPs enter the organism's body through three major routes: (a) ingestion (oral consumption) of contaminated food and water, (b) inhalation (breathing route) of air contaminated with MPs, and (c) skin contact with contaminated air, water, cosmetics.

Consumption of processed seafood products can cause health risk in humans. According to a global study, by the World Wide Fund for Nature, an average individual devours around 5 g of plastic particles weekly through fish diet (Chinglenthoiba et al., 2022). If dust on plates settling during suppertimes is taken into consideration, the estimated intake may be substantially greater. Bottled water intake accounts for approximately 118 to 325 MP particles/L. Therefore, consuming the suggested amount of water exclusively from bottled sources could result in ingesting 90,000 MPs annually (Mason et al., 2018). Airborne MPs on the surface of leafy vegetables were reported (He et al., 2023) and 100 nm MPs were difficult to remove even after washing. Additionally, studies show MPs presence in edible vegetables (carrot, lettuce, potato, broccoli) and fruits (apples, pears) which are more contaminated with <10 μm sized MPs (Conti et al., 2020). When plastic food packaging is exposed to high temperatures, it discharges over 10 million MP/mL of water, particularly PP and PET. Even recycled plastic causes consistent MP leaching, contaminating the food being packaged and raising the risk of food safety. Dietary intake of these toxins causes impairment of various human body systems as shown in (Fig. 7). To assess the threat of MPs to human health, it would be important to have knowledge of MP tissue accumulation and their consequences in mammalian models. The findings on MP-induced toxicity in several organ systems and their mechanism are given in the following subsections.

4.1 Digestive System

Digestive system is the main exposure route of MPs. The occurrence of PP and PET-MPs (50–500 μm) was observed in human stool, with a median of 20 MP particles/10 g of fecal matter (Schwabl et al., 2019). After intake, MPs may move through the tissue via paracellular transport, through endocytosis by M-cells and reach the gastrointestinal tract resulting in systemic exposure. Cytotoxicity of MPs was observed in three in vitro systems (goblet cells, Caco-2, and M-cells) and it increased with increasing concentration of PS-MPs. An in vivo investigation with mice was conducted, where no identifiable

Table 1 Microplastics adverse effects on aquatic organisms including microalgae and fishes

MPs	Size	Concentration MPs & contaminants	Aquatic organism	Affected Organ/toxicity	Effects	References
Toxicity on microalgae						
PS (green fluorescent)	1 µm & 5 µm	2, 10 & 50 mg/L	*Chlorella pyrenoidosa* (freshwater algae)	Molecular level	Inhibit growth, reduces photosynthetic pigment, alters transcript levels of the photosynthesis-related genes	(Cao et al., 2022)
PS	0.1 µm	25–200 mg/L	*Phaeodactylum tricornutum* (diatom)	Physiology	Lower the Fv/Fm, chlorophyll a content, and cell density, reduce growth	(Lang et al., 2022)
HDPE, PVC & PP	100 µm	5–250 mg/L	*Acutodesmus obliquus* (marine algae)		Photosynthetic efficiency (F_v/F_m) decreases	(Ansari et al., 2021)
PS	1 µm &100 nm	5 mg/L	*Microcystis aeruginosa* (freshwater cyanobacterium)		1 µm MPs promote algal growth, inhibit photosynthesis, increases content of intracellular microcystin	(D. Wu et al., 2021a, 2021b)
LDPE		25–175 mg/L	*Chaetoceros calcitrans* (diatom)		Inhibit growth rate, retard photosynthesis, ROS accumulation	(Senousy et al., 2023)
Toxicity on aquatic fishes						
LDPE	125–250 µm	Methylmercury, brominated flame retardants (BFRs), PCB	Zebrafish (*Danio rerio*)	Liver	Alter organ homeostasis, MPs alone does not produce relevant effects	(Rainieri et al., 2018)
LDPE, HDPE, PP & PS		0.01, 0.1 & 1%	Japanese Medaka (larvae)		Behavioral alteration (swimming), decrease head/body ratio, 0.1% MPs was most toxic	(Pannetier et al., 2020)
PVC		Triclosan & MP (0.26, 0.69 & 1.6 mg/dm^3)	*Artemia salina*	Neurotoxic	Decline in AChE activity	(Albendín et al., 2021)

Table 1 (continued)

MPs	Size	Concentration MPs & contaminants	Aquatic organism	Affected Organ/toxicity	Effects	References
PS	157 +/- 52 µm		Zebrafish	Development toxicity	Hypoxic condition in embryos induces delayed hatching, elevated blood flow velocity and heart rate	(Duan et al., 2020)
PS	0.5 & 50 µm	100 & 1000 µg/L	Zebrafish (adult male)	Gut	Gut inflammation and changes in microbiota diversity	(Jin et al., 2018)
PS	32–40 µm	Cadmium (50 µg/L) & MP (50 or 500 µg/L)	Discus fish (*Symphysodon aequifasciatus*)	Immune toxicity	No effect on growth, induce oxidative damage, stimulate immune response	(Wen et al., 2018a, 2018b)
PS, PET & PE	3 mm, PE (150–180)	PE (0.33 & 1.33 mg/L)	Rainbow trout (*Oncorhynchus mykiss*)	Yolk-sac larvae	No effect on larval survival, limited effect on growth; PE increases genotoxicity and cortisol levels, but PS and PET had no adverse effects	(Jakubowska et al., 2022)
PS	1 µm	1×10^6 and 1×10^7 items/L	Swordtail fish (*Xiphophorus helleri*)	Liver	Liver metabolism disorders	(Y.-K. Zhang et al., 2022a, 2022b, 2022c, 2022d, 2022e, 2022f)
Microfibers (MF) and microbeads (MB)	500 µm; 0.2 and 1.0 µm	10 and 100 fibers/L	Goldfish	Liver	Increased AST and ALT levels indicates liver damage; DNA damage and OS more in MB	(Kim et al., 2023)
PET	2 mm	0.25–1 mg/L	Planktivorous fish (*Acanthochromis polyacanthus*	GI tract	Negative effect on the growth	(Critchell & Hoogenboom, 2018, 2024)
PP		1, 10 and 100 mg/L	Zebrafish	Liver	ROS production, alteration in antioxidant parameters induces OS in liver, apoptosis in blood and increased AChE activity	(Priyadharshini et al., 2024)

Table 1 (continued)

MPs	Size	Concentration MPs & contaminants	Aquatic organism	Affected Organ/toxicity	Effects	References
PS	0.1 µm	Copper	Nile tilapia (*Oreochromis niloticus*)	Liver, intestine	Changes histology, Oxidative damage, inflammation, alter intestinal microbiota, impair immune ability	(F. Zhang et al., 2022a, 2022b, 2022c, 2022d, 2022e, 2022f)
PVC	100 to 200 µm	10%, 20% and 30%	*Cyprinus carpio* var. larvae	Spleen, liver, intestine, kidney	Hemosiderosis and vacuolar degeneration in the spleen, glomeruli tuft shrinkage, inflammatory cell infiltration, damaged intestinal villi, cytoplasmic vacuolation in the liver	(Liu et al., 2023)
PS	50 or 500 nm	0.1–10 mg/L	Zebrafish	Fin	Inhibited fin regeneration by altering immune response, signals regulating fin regeneration and ROS signaling	(Gu et al., 2020)
PS	5 and 50 µm	100 and 1000 µg/L	Zebrafish (larvae)		Altered microbiome, changed glycolipid and energy metabolism	(Wan et al., 2019)
PS	2–4 µm	Amitriptyline hydrochloride (2.5 µg/L): MP (440 µg/)	Zebrafish	Gut	Elevated ROS production, cracking of intestinal villi, microbiota dysbiosis and intestinal inflammation	(Shi et al., 2023)
PS	10 µm	tetracyclines (50 µg/L)	*Oryzias melastigma*	Gut	Inhibited body length, alter microbiota composition and liver metabolism via the gut-liver axis	(P. Zhao et al., 2023a, 2023b)

Table 1 (continued)

MPs	Size	Concentration MPs & contaminants	Aquatic organism	Affected Organ/toxicity	Effects	References
PE	40 to 48 µm	Copper (Cu^{2+})	*Litopenaeus vannamei*	Heptopancreas,	Disrupt immune and antioxidant system and hemocyte count; lowers survival rate	(Y.-T. Chen et al., 2024a, 2024b, 2024c, 2024d)
MP		Phenanthrene (8–10 µg/L), MP (200 µg/L)	*Oryzias melastigma*	Ovary	Damage HPG axis, reduced ovarian maturity and induce reproductive toxicity; induce transgenerational toxicity (high deformity rate, low heartbeat)	(Yuejiao Li et al., 2022a, 2022b, 2022c)
PET, PP	< 100 µm	5,50 mg/L	*Daphnia magna*	Reproductive organ	Decreased first brood and total number of offspring	(Daniel et al., 2024)
PS	5 µm	Arsenic (As)	Zebrafish (female)	Ovary	Increased E2/T ratio causes disturbed sex hormone level and, oocyte maturation; effect offspring development	(Luo et al., 2025)
PS	2-µm	Amitriptyline (2.5 µg/L), MP (0.44 mg/L)	Zebrafish	Behavioural response	Stronger behavioral toxicity alter locomotor activity and social interaction; decreased SOD, CAT, and GSH levels induces ocular oxidative damage	(Yi Zhang et al., 2023a, 2023b, 2023c, 2023d, 2023e)

PS polystyrene; *HDPE* high density polyethylene; *PVC* polyvinyl chloride; *PP* polypropylene; *LDPE* low density polypropylene; *ROS* reactive oxygen species; *PCB* polychlorinated biphenyl; *AChE* acetylcholinesterase; *PE* polyethylene; *PET* polyethylene terephthalate; *AST* aspartate aminotransferase; *ALT* alanine aminotransferase; *OS* oxidative stress; *SOD* superoxide dismutase; *CAT* catalase; *GSH* glutathione histological lesion or inflammatory response were observed despite being given a blend of PS-MPs (1, 4 and 10 µm) orally three times a week (Stock et al., 2019). These findings confirm that exposure to ambient levels of MP particularly by oral ingestion does not pose serious health risk to humans. Another study, showed that PS-MPs (0.1, 0.5, 1, and 5 µm) cause harm to human intestines using small intestinal epithelial cell (HIEC-6) and human colonic epithelial cell (CCD841CoN) treated for 24 h. When these cell lines were treated with higher concentration of MPs (0.1 µm), they affected intracellular ROS levels and induced OS which damaged the intestinal cells. MPs (5 µm) adsorbed tightly on the cell membrane and affected the mitochondrial membrane potential and membrane functionality (Y. Zhang et al., 2022a, 2022b, 2022c, 2022d, 2022e, 2022f). MPs also inhibited the growth of probiotics (*Lactobacillus*) in digestive system and disrupted the intestinal microbiota (H. Chen et al., 2022a, 2022b; Wu et al., 2025). Gastrointestinal digestion model (in vitro) was used to look into the impact of other kinds of MPs (PS, PVC, PET and PE) on human digestion. Results showed that PS-MPs even at lower concentration (80 mg/L) caused a reduction in lipid digestion and also reduced the lipase activity. This study revealed the possible mechanism for inhibition of lipid digestion by MPs. The bioavailability of lipid molecules for digestion is decreased when PS-MPs form heteroaggregates with lipid droplets due to hydrophobicity and adsorption. MPs alter the secondary structure and upset the crucial open conformation of lipids (Tan et al., 2020). Donor-dependent effects were brought on by recurrent exposure to PE-MPs (21 mg/day) for 14 days. An altered luminal and mucosal gut microbiota was observed using mucosal artificial colon (M-ARCOL) model and a co-culture of intestinal

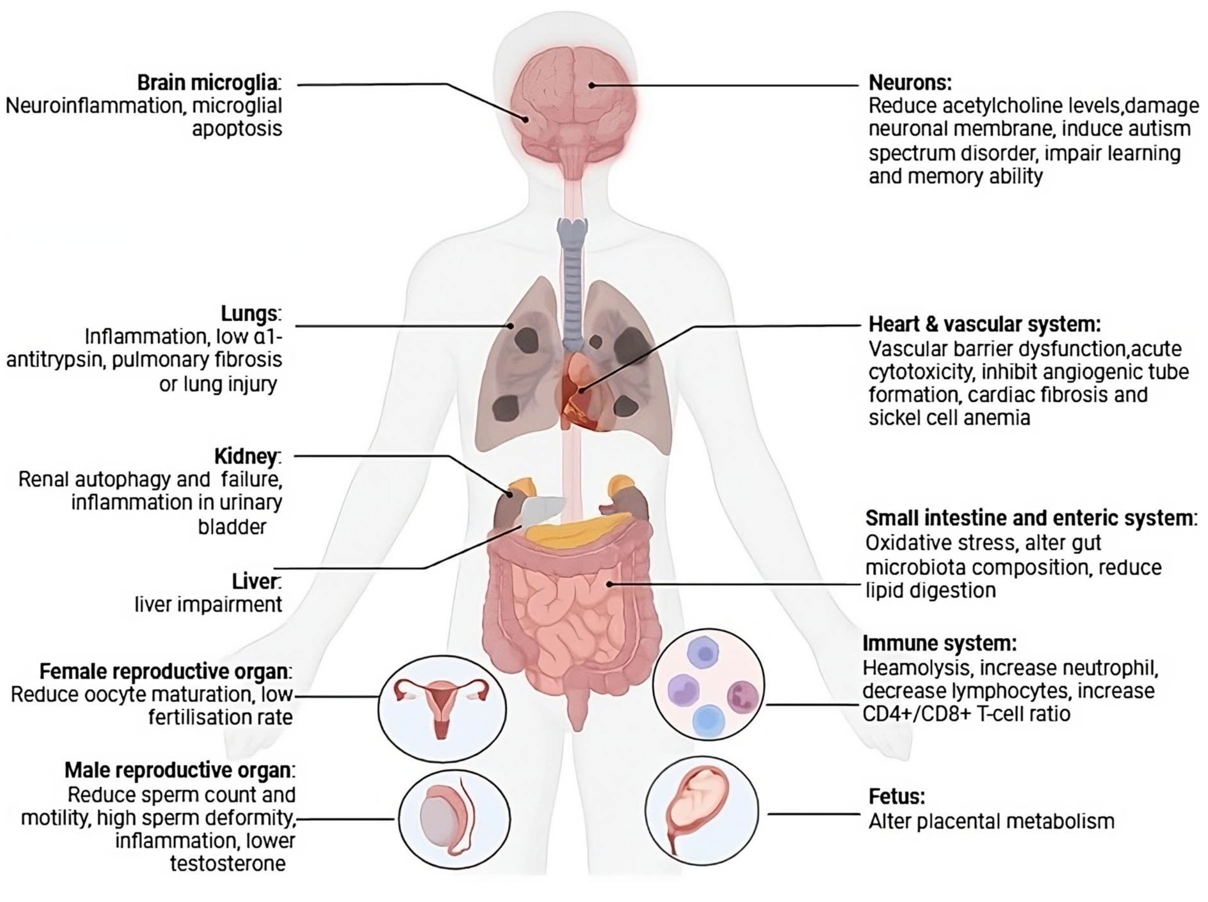


Fig. 7 Microplastics toxicity on the different organs of the human body

epithelial and mucus-secreting cells. In terms of microbiota perturbations: abundance of *Desulfovibrionaceae* and *Enterobacteriaceae*, dangerous pathobionts increased and while that of helpful bacteria viz *Christensenellaceae* and *Akkermansiaceae* decreased (Fournier et al., 2023). But there was no evidence of a major effect of PE-MPs on the intestinal barrier related to gut microbial metabolites. A recent study by (Prabhu et al., 2024) provided the evidence for interaction between MPs (PS and PP) and digestive enzymes mainly duodenal and bile juices. High roughness of MPs surface was corelated to MP-enzyme dehydrogenation reactions and NO addition that result in corona formation. This causes mass fragment loss, changes in morphology, and functionalization of MPs. Also, observed was leaching from MPs during digestion process may accumulate in different digestive organs thus imparting toxicity. An investigation revealed the interaction of small sized PS-MPs with the digestive enzyme i.e. pepsin by tracking the rates of protein hydrolysis through static in vitro gastric digestion of MP contaminated cow's milk. The $\pi - \pi$ interactions between the aromatic residues and the PS phenyl rings appear to be the main factor in pepsin adsorption. Prolonged exposure of MPs causes structural changes in protein thereby reducing its activity. In the presence of small PS, digestion resulted in decreased bioavailability of short peptides (2–9 kDa) in the gastric phase and a temporary buildup of larger peptides of cow's milk proteins (10–35 kDa) in the digestion mixture (de Guzman et al., 2023). (D. Zhang et al., 2024a, 2024b, 2024c) reported the presence of MPs in the gall stone and further investigated that MPs have higher affinity for cholesterol. Using mice model, they concluded that MPs aggravate cholelithiasis by forming MP-cholesterol heteroaggregate and also altered the gut microbiota. Thus, the studies confirmed that the MPs have the capability to interact and alter the digestion processes such as lipid and protein digestion and also the digestive enzymes.

4.2 Respiratory System

Inhalation is another major route of exposure for MPs. Aerial MPs have detrimental effects on the human respiratory system. These MPs can traverse within the atmosphere over 95 km. MPs were also reported in upper respiratory tract mainly in nasal lavage and sputum of workers (Jiang et al., 2022). The occurrence of airborne MPs (12 polymer types) was observed in lung tissues comprising mainly of PET, resin, and PP fibers. The sizes ranged between 8.12 and 16.8 µm (Amato-Lourenço et al., 2021). Similarly, inhaled PS-MPs caused pulmonary fibrosis as confirmed by the Masson staining. MP administration upregulated the β-catenin and TCF4 levels as assessed by the immunofluorescence and western blot. The latter also indicated the activation of Wnt/β-catenin signaling pathway, a necessary pathway related to pulmonary fibrosis. Also, suppressed the levels of GSH and SOD were observed which are indicative of OS in lungs (X. Li et al., 2022a, 2022b, 2022c). Human lung epithelial cells (BEAS-2B) evidenced pulmonary toxicity upon PS-MP exposure. High doses of MP induce the formation of ROS that increases IL-6 and IL-8 thereby causing cytotoxicity and inflammatory effects. Also, damage to zonula occludens (ZO) proteins was noted indicating impaired pulmonary barrier (Dong et al., 2020). In vivo and in vitro studies confirmed that PS-MPs cause respiratory diseases in a size dependent manner with smaller ones causing more severe toxicity. Transfection experiments show that smaller PS-MPs (1–5 µm) target TLR2 receptor, induce high levels of inflammatory responses and lead to overexpression of TLR2. There was an induction of OS, apoptosis, and exacerbated fibrosis by activation of NF-κB signaling (Cao et al., 2023a, 2023b). Moreover, the negative impact of airborne MPs to human health was tested in mice with both asthmatic and normal pulmonary physiology. In normal mice, MPs cause inflammatory cell infiltration, high TNF-α levels in bronchoalveolar lavage fluid (BALF), with high IgG1 production and macrophage aggregation and the same was observed in asthmatic one (Lu et al., 2021). Pulmonary toxicity was also reported due to inhalation of tire wear MPs (TWMPs) leading to pulmonary fibrotic lung injury. This is caused by earmarking the cytoskeletal regulatory proteins twinfilin-1, miR-1a-3p which impede the formation of F-actin and impact the cytoskeleton rearrangement (Yanting Li et al., 2022a, 2022b, 2022c). In (Kwabena Danso et al., 2024) study, mice (C57BL/6) received intratracheal injections of 5 mg/kg PP, PS, and PE every day for 14 days to examine pulmonary toxicity. Also, there was an increased expression of TLR4 in the lung tissues. This induced activation of NF-κB and NLRP3 inflammasome that ultimately induce pulmonary inflammation. These studies show MPs can persist in lung tissues and exacerbate pulmonary toxicity by inducing oxidative damage and apoptosis via NF-κB and Wnt/β-catenin signaling pathway.

4.3 Vascular System

Recently, some studies evidenced MPs in blood stream of healthy people. (Y.-C. Chen et al., 2023a, 2023b) researched PS-MPs toxicity in human circulatory system using endothelial EA. hy926 cells. MPs can cause oxidative damage which potentially leads to apoptotic cytotoxicity. PS-MPs have the ability to reduce the protective potential of heat shock proteins. Furthermore, through the depletion of ZO-1 protein, PS-MPs cause vascular barrier dysfunction. PS-MPs (460 nm and 1 μm) at high concentration cause hemolysis due to their adherence to RBCs via weak van der Waals forces. Also, an increased expression of IL-6 indicative of early stage inflammation was observed (Hwang et al., 2020). (R. Sun et al., 2021a, 2021b) studied the PS-MPs toxicity (5 μm) in mice hematological system by performing colony-forming (CFU) assay and peripheral blood routine. The result showed lower counts of WBCs and lower colony counts of granulocyte–macrophage progenitor cells CFU-GM, CFU-G, and CFU-M. Thus, bone marrow (BM) cells' capacity to form colonies was impeded. This study revealed that exposure to PS-MPs have an impact on peripheral blood cell counts by inhibiting BM hematopoietic stem cells from differentiating into granulocytes and megakaryocytes. This knowledge about hematotoxicity in BM cells of terrestrial mammals will assist in determining risk to health. PS-MPs (1 μm) induce acute cytotoxicity to vascular endothelial cells as evidenced by experiments on human umbilical vein endothelial cells (HUVECs) (Y. y. Lu et al., 2023). Further (Lee et al., 2021) examined the effect of 0.5 μm PS-MPs on HUVECs showing endothelial dysfunction and anomalous vasculature. Exposure for 6 h showed inhibited angiogenic tube formation by suppressing vascular endothelial growth factor (VGEF) signaling pathway while prolonged exposure lowered the cell viability and caused autophagy and necrosis-mediated cytotoxicity. (Z. Li et al., 2020a, 2020b) reported the toxic effect of PS-MPs on cardiovascular system. Exposure for 90 days, showed raised levels of creatine kinase-MB (CK-MB) and Troponin I in serum which caused structural impairment of myocardium. It resulted in induction of OS by stimulating the Wnt/β-catenin signaling pathway that causes cardiac fibrosis ultimately leads to cardiac dysfunction. When 600 μg/day MPs (high concentration) were administered for 15 days; an alteration of RBCs structure which is an indication of sickle cell anemia and altered biochemical parameters that showed renal and liver impairment were observed (Abdel-Zaher et al., 2023). This study concluded with thrombin-fibrinogen clot model that the ultimate turbidity and the rate at which thrombin converts fibrinogen are both lowered by the presence of MPs. Thus, inhibiting the formation of fibrin clots based on the surface charge and concentration (Tran et al., 2022). According to the current study (Lee et al., 2024), MPs are identified in human blood and have a favorable correlation with a number of lifestyle factors as well as significant alterations in coagulation indicators. The coexistence of MPs and Pb in human samples was evaluated in aortic disease. Further in vitro study evidenced the pathway by which excessive ROS formation disturbs mitochondrial function and led to inflammation and PANoptosis (pyroptosis, apoptosis and necroptosis which form the death triangle of cells) through cAMP/PKA/ROS signaling pathway. This encouraged the release of inflammatory chemicals, degeneration of vascular smooth muscle cells (VSMCs), and the degradation of elastic fibers (Xie et al., 2024). Prolonged exposure to MPs induces hematotoxicity. They are circulated via the circulatory system to other organs causing cardiac dysfunction.

4.4 Reproductive System

After entering the bloodstream, MPs deposit in the testicles, ovaries, and placenta after passing through the blood-testis or placenta barriers. For the first time, (Q. Zhao et al., 2023a, 2023b) revealed the presence of different kinds of MPs (PS, PE, and PVC) in both testis, semen and ovarian follicular fluid (Montano et al., 2025). Thus, MPs can cause reproductive dysfunction as reported. PS-MPs in female mice induce reproductive toxicity upon 35 days persistent exposure. (Z. Liu et al., 2022a, 2022b), revealed that MP exposed organs accumulated PS-MPs, ovaries had raised IL-6 levels and reduced MDA levels. This indicated that the exposure to PS-MPs decreased the level of endoplasmic reticulum calcium ([Ca^{2+}] ER), mitochondrial membrane potential (MMP), GSH and increased ROS in oocytes. Therefore, MPs induced decline of oocyte quality and maturation in female mammals. Another study, (Yingbing Zhang et al., 2023a, 2023b, 2023c, 2023d, 2023e) reported toxicity to female reproductive system of mice post oral administration of MPs (30 mg/kg/day) for 30 days. Here, lowered oocyte maturation, fertilization rate, and embryo development were reported. Also, observed were elevated levels of ROS indicating an OS induction in oocyte and embryo. Also, observed were abnormalities in spindle and chromosomal shape. Actin and Juno expression was downregulated in them causing DNA damage in oocytes. Mice were additionally fed with MPs during lactation and gestation period to study the transgenerational reproductive toxicity. There was a reduction in birth, postnatal body weight of offspring and in survival rate. It is interesting to know that just as the mother, female offspring were also born with reduced oocyte maturation and fertilization rate. PS-MPs induce reproductive toxicity including increased sperm deformity mainly in spermatogenesis which is related to Nrf2/HO-1/NF-κB pathway (Hou et al., 2021). Similarly, (Xie et al., 2020) investigated that impact of PS particles on sperm motility and count, showing increased deformity rate and decreased activity of sperm metabolic enzymes. MPs exposed mice had lower serum testosterone contents and high OS as revealed by activation of p38 MAPK and JNK pathway resulting in reproductive toxicity. p38 MAPK pathway got inhibited by ROS scavenger N-acetylcysteine (NAC), and by p38 MAPK-specific inhibitor (SB203580) both of which improved the testosterone secretion. A study by (Qin et al., 2024) showed that the invasion of the uterus by MPs was controlled by the vagina-uterine lacuna systems (large sized particles) or diet-blood circulation (small sized particles). In F0 female mice, the lactation-induced release of PS-MPs from plastic feeding bottles during infant formula preparation resulted in reproductive damage. Female offspring did not exhibit reproductive disorders while male offspring showed transgenerational toxicity such as disrupted spermatogenesis that result in low sperm count and viability (Dou et al., 2024). The findings by (Wen et al., 2023) revealed that mice exposed to PS-MPs had more sperm abnormalities and lower sperm counts and motility. The proposed mechanism is that the MPs trigger IRE1α/XBP1s signaling pathway and downregulate apoptotic modulator (*Caspase12*,*9*,*3*), steroidogenic acute regulatory protein (*StAR*) and thus reduced testosterone synthesis. Others reported spermatogenic dysfunction via p53 signaling pathway (C. Lu et al., 2023); LHR/cAMP/PKA/StAR pathway (Jin et al., 2022a, 2022b) and JNK/p38 MAPK pathway (Xie et al., 2020). MPs induce reproductive toxicity in both male and female mice through different pathways and interestingly even the male offsprings show transgenerational toxicity.

4.5 Immune System

MPs exhibit immunotoxicity. Organisms depend on their immune systems to fight off MP stress. PP particles (~ 20 μm) showed cytotoxicity in size and dose dependent manner. They increased ROS, histamines and cytokines levels thereby stimulating immune system and causing hypersensitivity (Hwang et al., 2019). Another study revealed that MPs suppressed the immune function and damaged the spleen in mice by lowering expression of S100A8. All mice of MP treated group were given water with 5 μg/ml microbeads (5 μm) for 4 weeks. The results suggested that drinking water containing MPs can cause tissue damage and influence immunity by increasing $CD4^+$/$CD8^+$ T-cell ratio (Juan et al., 2022). MPs induced immunotoxicity as revealed by giving PE-MPs supplemented diet to mice for 90 days. An increased neutrophil levels while a decreased lymphocytes proportion at 2 mg dose/day was reported. IgA levels in blood stream and ratio of $CD4^+$/$CD8^+$ T-cell enhanced in a dose dependent manner. Migration of granules to mast cell membrane was observed in the stomach (Park et al., 2020). Further, PS-MPs exposure caused immunological disruption and adverse pregnancy outcomes (Hu et al., 2021). Chronic exposure to PP-MPs may cause immunological disruptions, particularly in females, leading to a predominance of type-2 helper T cell reactivity (Kusma et al., 2024). The findings by (Shang et al., 2024) demonstrated that mice exposed to PS-MPs during lactation period had higher spleen weights and increased B and regulatory T cells causing spleen immune dysfunction. Additionally, F1 male offspring also had enlarged spleen and elevated helper B and T cells but F2 generation had minimal changes in the spleen suggesting that it is not passed down through the generations. Thus, MPs depending on the concentration can cause immunotoxicity in humans even during lactation period and transgenerational effects in male offsprings.

4.6 Nervous System

Limited reports have shown the interaction of MPs and central nervous system (CNS) and their potential to cause neurotoxicity. MPs have the ability to penetrate the blood–brain barrier (BBB) and may harm the CNS of humans. MPs cause neuroinflammation and OS, which can result in neurodegenerative diseases (Moiniafshari et al., 2025). PS-MPs agglomerate in microglial cells of the brain (Kwon et al., 2022) and; human olfactory bulb (Amato-Lourenço et al., 2024) were observed. (Hua et al., 2022) revealed the adverse impact of PS-MPs on forebrain cerebral spheroids and found out that the smaller MPs adversely affect the development of brain tissues and, promote proliferation while higher concentration of MPs showed reduced cell viability and altered the neural gene expression. In addition, LDPE-MPs were found to be distributed in blood stream and reached hippocampal neurons afflicting them. They reduced the expression of neuronal SOD enzyme, decreased Aβ42 levels in blood serum and thus damaged the neuronal membrane (Sincihu et al., 2023). There are some evidences showing the adverse effect of PE-MPs inducing autism spectrum disorder (ASD) wherein MPs accumulate in the brain and brought behavioral changes (Zaheer et al., 2022). Moreover, PS-MPs (0.01, 0.1, 1 mg/d) when given orally to Kunming mice for 4 weeks have certain effects on learning and memory abilities which were impaired and were observed by the morris water maze test. This was due to disordered nerve cells in hippocampus region and reduced levels of ACh that caused OS ultimately inhibiting the pCREB pathway. Increased ROS levels and decreased GSH levels were also observed in brain tissues. It was interesting to know that after treating the cells with vitamin E, they regained their learning and memory power (S. Wang et al., 2022a, 2022b, 2022c).

4.7 Excretory System

MPs after ingestion get accumulated in kidney showing heightened danger of renal diseases. MPs can attach to the cell membrane and get engulfed by the cell. (Y.-C. Chen et al., 2022a, 2022b) reported nephrotoxicity due to consumption of PS particles experimenting on human embryonic kidney 293 (HEK293) cells. This association of MPs inhibits the antioxidant heme oxygenase-1 which induces OS. At lower concentration of MPs (3 ng/ml) inflammation is induced whereas higher doses (300 ng/ml) cause autophagy thus, lowering the expression of NLRP3. A lowered NLRP3 expression reduces the ZO-2 proteins and α1-antitrypsin, which compromise the integrity of the renal barrier and raised the risk of acute kidney damage. Another study, (Xiong et al., 2023) reported kidney injury in response to MPs of different sizes which ultimately promoted kidney fibrosis. It also altered the gene expression related to circadian rhythm and immune system. Furthermore, PS-MPs; when exposed to human kidney proximal tubular epithelial cells (HK-2 cells) showed high ER stress and increased Beclin 1 and LC3 levels i.e. autophagy related proteins. They altered the proteins involved in phosphorylation of AKT/mTOR and MAPK signaling pathways. Also, in vivo study conducted on male mice by giving MPs orally; an accumulation was observed which resulted in lesions and other detrimental effects on kidney (Y.-L. Wang et al., 2021a, 2021b). (Sun et al., 2023) revealed the kidney toxicity by the combined effect of DEHP and PS-MPs. PS-MPs promoted OS in the cells and switched on the AMPK/ULK1 pathway which leads to renal autophagy. Further, PS-MPs exposure causes necroptosis and inflammation of urinary bladder tissues which was studied for the first time (Y. Wang et al., 2022a, 2022b, 2022c). On PS-MP exposure to human kidney organoids showed accumulation in glomerular like structures and reduced the organoid size. MPs elevated ROS formation and induced apoptosis in nephron progenitor cell during early kidney development. MPs also reduced the proximal and distal tubular structure via Notch signaling thereby disrupting normal nephron and kidney development (Zhou et al., 2024). (Liang et al., 2024) revealed that

Table 2 Microplastics toxicity in vitro and in vivo systems

Human cells/model	MPs (size)	Concentration	Toxicity/route	Detailed effects	Result	References
MPs toxicity using human cell lines (in vitro studies)						
Caco-2 cells	PS & BPA (300 nm, 500 nm, 1 µm, 3 µm & 6 µm)		Cytotoxicity (ingestion)	Mitochondrial depolarization and elevated OS	Size and concentration dependent toxicity	(Q. Wang et al., 2020a, 2020b)
Human Coronary Artery Smooth Muscle Cells (HCASMCs)	PE (632 µm); PS (564 µm)	1 mg/mL	Vascular	Caused apoptosis and triggered pathological alterations including changes in migration and proliferation	Lead to cardiovascular diseases	(Persiani et al., 2025)
HEK 293 cells and human hepatocellular (Hep G2) liver cells	PS (1 µm)	100 µg/mL	Cytotoxicity (ingestion)	Increased ROS levels, inhibited cell proliferation but not the number of viable cells, and altered metabolism	Higher dose induces toxicity	(Goodman et al., 2022)
Human peripheral lymphocytes	PE (10–45 mm)	25, 50, 100, 250, and 500 mg/mL	Genotoxic	Genomic instabilities, elevated nucleoplasmic bridge formation and nuclear bud formation	Lower doses of MPs increase the level of genomic instability	(Çobanoğlu et al., 2021)
Human alveolar A549 cells	PS (1 & 10 µm)		Pulmonary (inhalation)	More uptake of 1 µm, changes the cell morphology, and decrease proliferation rate	The cell viabilities 93% maintained even at high concentration	(Goodman et al., 2021)
Human intestinal cell line HT29	PS (3 & 10 µm)	100–1600 particles/mL	Cytotoxicity	Increased ROS, and cell mortality rate	Acute effects causing damage to the acid compartment	(Visalli et al., 2021)
L929 murine fibroblasts and MDCK epithelial cell lines	PE (1.0–4.0 µm), PS microspheres (9.5–11.5 µm)	1, 10 & 20 µg/mL	Cytotoxicity	Altered transcriptional expression of antioxidant enzymes and inflammatory cytokines, metabolism and cell viability	Cell viability depends on dosage	(Palaniappan et al., 2022)

Table 2 (continued)

Human cells/model	MPs (size)	Concentration	Toxicity/route	Detailed effects	Result	References
Human colon adenocarcinoma Caco-2 cells	PS (0.1 μm & 5 μm)	1–200 μg/mL	Gastrointestinal cells	Disrupted mitochondrial membrane potential, inhibited ABC transporter activity in the plasma membrane and elevated arsenic toxicity	Large sized MPs induce high toxicity	(Wu et al., 2019)
Human cell Caco-2	PE & TBBPA (30 to 140 μm)	100, and 1000 mg/L	Gut	Weakened cell vitality, altered human gut microbiota's composition, inhibited metabolic pathways, and breaks intestinal homeostasis	At high concentrations MPs increase TBBPA's cytotoxicity	(Huang et al., 2021)
Normal human liver cells, THLE-2,	PS (0.1 and 1 μm)	10 μg/mL	Liver	Disturbed the metabolic profile, and altered ABC transporter pathway	Hepatoxicity induced by chronic exposure	(Z. Chen et al., 2024a, 2024b, 2024c, 2024d)
Skin squamous cell carcinoma cell lines (A431 and SCL-1)	PE (1 μm)	0–1 mg/mL		Promoted the proliferation of tumor cells and damage to normal skin cells	Skin cancer	(Y. Wang et al., 2023a, 2023b, 2023c)
Human dermal fibroblast-derived spheroids	PS (0.1, 0.5, 1, and 3 μm)		Skin	Morphological alterations, and affect gene expression linked to ECM	Affect dermal system	(Eom et al., 2024)
MPs toxicity in vivo studies						
Mouse & human microglial HMC-3 cells	PS (0.2,2 and 10 μm)		Brain (7 days oral treatment)	Deposited in brain (microglial cells), and changed immune response gene expression	Microglial apoptosis	(Kwon et al., 2022)
Pietrain X Duroc breed gilts	PET (300 μm)	0.1 –1 g/animal/day	Digestive tract	Altered enteric nervous system and duodenum's histological structure; histopathological changes	Higher concentration of MPs had more influence	(D. Zhang et al., 2024a, 2024b, 2024c)

Table 2 (continued)

Human cells/model	MPs (size)	Concentration	Toxicity/route	Detailed effects	Result	References
Rat (5-week-old)	PS (0.3 μm)		Cardiovascular	Mean BP increased by 22–40% via bradykinin-NO pathway; cause heart oxidative damage and cardiomyocyte hypertrophy via ERK activation	Varied effect of PS-MPs with different surface functional groups	(Du et al., 2025)
BALB/c mice (6-week-old)	0.5, 4, and 10 μm	100 & 1,000 μg/L	Brain (180 days exposure)	BBB disruption and inflammatory response; learning and memory dysfunction	Cognitive impairment and neurotoxicity	(Jin et al., 2022a, 2022b)
ICR male mice (5-week-old)	PS (5 μm)	100 μg/L, 1000 μg/L & 10 mg/L	Testis (35 days exposure)	Reduced quantity of viable epididymis sperm, elevated sperm deformity rate and inflammation in testicular tissues	Abnormal spermatogenesis	(Hou et al., 2021)
Mice	PS (50 nm & 2 μm)		Ocular surface (2–4 weeks)	Deposited in lower eyelid (conjunctival sac), goblet cells number decreased, dry eye, and inflammation of the lacrimal gland and conjunctiva	Ocular surface dysfunctions	(Zhou et al., 2022)
Drosophila melanogaster (third–instar larvae)	PS (4, 10, or 20 μm)	0.01–10 mM	Genotoxic (ingestion)	Defective morphology, somatic DNA recombination in the wing imaginal disk cells	Produced genotoxic effects in a concentration dependent manner	(Demir, 2021)
ICR mice	PE (10–50 μm)	500, 1000, and 2000 mg/kg/day	(28-days oral treatment)	No changes in body weight, and inflammation in lung tissues	Lung damage	(S. Lee et al., 2022a, 2022b)
Mice	PE (30 and 200 μm) & arsenic	2, 20, and 200 μg/g	Oral	Up-regulated gut metabolite expression	Affect gut microbiota dose dependently	(S. Chen et al., 2023a, 2023b)
C57BL/6-mated BALB/c mice (8–10 weeks old)	PS (10 μm)		Intraperitoneally injected (reproductive)	Reduce the uterine blood supply, immune disturbance, and elevated embryo resorption rate	Effects on pregnancy outcomes	(Hu et al., 2021)

Table 2 (continued)

Human cells/model	MPs (size)	Concentration	Toxicity/route	Detailed effects	Result	References
BALB/C mice (6-week-old)	PS (0.5 µm, 4 µm, 10 µm)			Testosterone levels and sperm quality decreased; disturb blood-testis barrier and induce testicular inflammation	Male reproductive dysfunctions	(Jin et al., 2021)
NMRI male mice (6 weeks old)	PS (2 µm)		6 weeks treatment	Reduced sperm count and motility; linked with OS and ER stress	Tissue damage in testis even at low doses (0.01 mg/kg)	(Zangene et al., 2024)
C57BL/6 female mice (7 to 12 weeks old)	PE microbeads (36 & 116 µm)	100 µg/g	Food treatment (6 or 9 weeks)	Increased cytokine expression and inflammatory foci, fostered oxidative imbalance, and improved liver proliferation	Liver fibrogenesis	(Djouina et al., 2023)
Three-dimensional CO model & male Balb/c mice	PS (1 µm)	0.025, 0.25 and 2.5 µg/mL	Instilling intratracheally (4 weeks)	Elevated levels of apoptosis, OS, inflammatory response, and collagen accumulation and increases in interventricular septal thickness	Induced cardiac hypertrophy	(Y. Zhou et al., 2023a, 2023b)
Sprague Dawley rats	PS (100 nm, 500 nm, 1 µm and 2.5 µm)	0, 0.5, 1 and 2 mg/200 µL	Intra-tracheal instillation (2 weeks)	1 µm deposited in lungs; downregulated circRNAs, upregulated lncRNAs and upregulated pro-inflammatory cytokines induce damage alveolar	Induce lung inflammation and injury	(Fan et al., 2022)

Table 2 (continued)

Human cells/model	MPs (size)	Concentration	Toxicity/route	Detailed effects	Result	References
C57BL/6 J mice	PS (5 µm)	(0.6, 3, and 15 mg/kg)	Intratracheal instillation (60 days)	Increased gram-negative bacteria cause LPS release; TLR4 activated and ferroptosis occurs induce homeostatic imbalance; increase collagen fibers and decrease membrane permeability	Lung injury	(Kang et al., 2024)
ICR mice	PET (10 µm)	2000 mg/kg	Oral administration (1 week)	Concentration dependent granulomatous inflammation	Chronic inflammatory effects	(D. Kim et al., 2025a, 2025b)
Renal interstitial fibroblasts (NRK-49F cells)	PS		Oral given (6 months)	Inflammation and fibrosis; induce ferroptosis via ferritinophagy and secreted TGF-β1	Renal fibrosis	(Hong et al., 2024)

PS polystyrene; *OS* oxidative stress; *BPA* bisphenol A; *ROS* reactive oxygen species; *PE* polyethylene; *MDCK* Madin–Darby canine kidney; *HEK* Human embryonic kidney; *ABC* ATP-binding cassette; *TBBPA* Tetrabromobisphenol A; *ECM* extracellular matrix; *PET* polyethylene terephthalate; *BP* blood pressure; *NO* nitric oxide; *BBB* blood brain barrier; *ER* endoplasmic reticulum; *CO* cardiac organoid; *LPS* lipopolysaccharide; *TLR* toll-like receptor

PS-MPs cause disruption of the gut barrier by elevating the urinary complement-activated product C5a and renal C5a/C5aR expression which induces kidney damage or injury.

4.8 Other Adverse Effects

MPs can transport easily to different organs and tissues to accumulate and exert potential toxicity. (Pan et al., 2023) have provided evidence of MPs interaction with the skeletal system. They reported accumulation of PS-MPs in axial and long bones causing lowered femur and tibia length. MPs also lower the osteogenic ability and hasten the osteoblast senescence thus arresting the skeletal growth in puberty. A study showed that MPs are widely present in human joints by examining synovial samples of patients having knee or hip replacements. There were nine distinct polymer types found and a stronger cellular stress response was linked to elevated MP levels (Z. Li et al., 2024a, 2024b, 2024c). BPA (bisphenol A) and PS-MPs when combined caused hepatotoxicity (Cheng et al., 2023). Interestingly, PS-MPs can be transported from pregnant mother to fetus and are responsible for altering the placental metabolism (Aghaei et al., 2022). Research indicates that MPs may interfere with hepatic enzyme function and damage DNA, which may result in cancer. Results showed that teabags released high amount of DEHP into tea beverages which promotes cancer risk (CR) in children and adults (Kashfi et al., 2023). MPs and other contaminants cause toxicity at certain concentration and affect the different body systems as described in Table 2. Several studies both in vitro and in vivo have shown the detrimental effect on human health.

5 Conclusion and Future Prospects

The review addresses omnipresence of MPs in the environment and their aggressive reach into animals, humans, terrestrial and aquatic flora. Besides, their aggravated use, improper management and recycling of MPs has led to serious repercussions. The adverse effects in plants and planktons range from low photosynthetic productivity, low nutrient density along with disruption of soil microbiota. In animals, they cause neurotoxicity, inflammation, cancers, gastrointestinal and oxidative stress and, transgenerational reproductive toxicity in both male and female mice. The review compiles a thorough peep into source, abundance and MP ecotoxicology on environment and human health. From planktons to edible plants and further higher organisms have observed a cascading MP prevalence due to trophic transfer. An increase in global plastic footprint and NCDs is observed which may have an underlying correlation. However, a global such as this can only be resolved via multidirectional and interdisciplinary one health approaches. Scientists working on climate change and pollutants along with biologists are required to work in concert to test highly polluted environments, also geospatially map them and then devise a strategy in concert to eradicate MPs. Despite advancements in our understanding of MPs, a startling dearth of pertinent data persists which should be addressed at a global scale. Attention should be paid to the information on the bio-ecological effects and trophic transfer of MPs in freshwater organisms. Research and development of new bioplastics and biodegradable plastics should be focused on. Besides this, the efforts must continue in direction of reducing plastic consumption, improving waste management and reclaiming contaminated soil and water by biodegradative approaches using microbes.

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Data Availability The data for Fig. 1. are available at https://ourworldindata.org/plastic-pollution and https://ourworldindata.org/grapher/share-of-total-disease-burden-by-cause

Declarations

Ethics Approval The study requires no ethical approval. There are no human subjects involved.

Consent to Participate There was no consent to participation or ethical approval required for the manuscript.

Consent to Publish WE, the authors of the manuscript **Multi-dimensional view microplastic toxicity: Environmental and Health impact** consent to publish the manuscript to the journal Water, Air, & Soil Pollution.

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The authors declare the following financial interests/personal relationships which may be considered as potential competing interests.

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